

BIOL3005

Cognition

Answers to the self-check questions

A companion to Cognition: An open textbook

Welber Marinovic

Curtin University, Perth, Western Australia

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About these answers

This companion gives model answers to the self-check questions at the end of each chapter of *Cognition: An open textbook*, written for BIOL3005 Cognition at Curtin University. The answers are organised by chapter and numbered to match the questions in the textbook, and each question is reproduced above its answer.

Each answer sets out what a complete response covers, using the evidence, studies and terms that the chapter itself uses. They are not the only acceptable answers. A different answer that is accurate, uses the evidence well and reasons carefully can be just as good, and for the more open questions, such as those asking you to design a study or a revision plan, there are many good answers.

Try each question yourself before reading the answer. Retrieving an answer from memory, even an incomplete one, does far more for long-term retention than reading a correct answer (Chapter 5), so these answers work best as a way of checking and extending your own attempt. References to sections (for example, §4.4) are to the textbook, and the full references for every study mentioned are in the relevant chapter of the textbook.

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Chapter 1: Introduction & Cognitive Neuroscience

Question 1

Explain how Donders used the difference between choice and simple reaction time to estimate the duration of a decision. What assumption does the subtractive method rely on?

Donders measured simple reaction time (press one button when a light appears), go/no-go reaction time (respond to one designated stimulus and withhold the response to others) and choice reaction time (a different button for each light). Each task contains the operations of the one before plus one more: go/no-go adds identifying the stimulus, and choice adds selecting which response to make. Subtracting one mean reaction time from another therefore estimates the duration of the added operation. Choice minus simple gives the combined time to identify the stimulus and select a response, which Donders treated as the time to decide, about 100 ms; choice minus go/no-go isolates response selection alone. The method rests on the assumption of pure insertion: adding a stage leaves the other stages unchanged, so the components simply add up. If inserting an operation altered how the shared stages ran, the difference in reaction times would no longer measure the inserted stage alone.

Question 2

Why did introspection fail as a method, and how does that failure connect to the claim that most cognition is unconscious?

Introspection asked trained observers to report the elements of their conscious experience. Even under Wundt's strict conditions (practised observers, repeatable stimuli) observers never reached agreement, and because the reports were subjective there was no way to check from outside whether anyone was describing their mental contents accurately. The deeper problem is that most cognition is unconscious. The regulation of breathing, the filtering of background sound and the holding of the previous sentence in mind all go on without awareness, so introspection can only ever reach the small fraction of processing that surfaces into consciousness. A method that can only see the rider will always have a low ceiling for explaining a mind that is mostly elephant.

Question 3

What is a double dissociation, and why is the Broca–Wernicke pairing stronger evidence for localisation than a single case of aphasia would be?

A double dissociation is a pair of complementary findings: damage to one region impairs function A but spares function B, while damage to another region impairs B but spares A. Damage to Broca's area in the frontal lobe impairs speech production while leaving comprehension of everyday language largely intact; damage to Wernicke's area in the temporal lobe impairs comprehension while leaving speech fluent, though empty of meaning. A single case of aphasia shows only that one ability can be lost while another survives, which could happen simply because one task is harder or more fragile than the other. When each ability can be lost while the other is spared, that explanation fails: the two cannot be one function riding on a shared mechanism, so they are separable and depend on different regions.

Question 4

A split-brain patient cannot name an object flashed to the left visual field but can pick it out with the left hand. Explain why, in terms of hemispheric wiring and lateralisation.

Each visual field projects to the opposite hemisphere, so an object flashed to the left visual field reaches only the right hemisphere. In most people language is lateralised to the left hemisphere. With the corpus callosum cut, the right hemisphere cannot pass the information across, so the verbal left hemisphere never receives it and the patient cannot name the object. The right hemisphere has nonetheless seen it, and because each hemisphere controls the opposite side of the body, it can guide the left hand to pick the object out by touch. Flash the same

object to the right visual field and it reaches the language-dominant left hemisphere, so the patient names it easily. In an intact brain the two hemispheres work together, so the split should not be overstated.

Question 5

Distinguish episodic, semantic, and procedural memory, and give an everyday example of each.

Episodic memory is memory for personally experienced events, with their what, where and when, such as remembering your first lecture in this unit and where you sat. Semantic memory is factual knowledge detached from the occasion on which it was learned, such as knowing that Perth is the capital of Western Australia without remembering when you learned it. Both are explicit (declarative) because they can be consciously brought to mind. Procedural memory is implicit (non-declarative) memory for skilled actions, expressed in performance rather than recollection, such as riding a bike, swimming freestyle or typing without looking at the keys.

Question 6

Why does random assignment license a causal inference, and what is a confound? Explain how a double-blind procedure protects against two specific confounds.

When participants are randomly assigned, they do not choose their condition, so on average the groups are equivalent on everything except the manipulated independent variable. Any reliable difference in the dependent variable can then be attributed to the manipulation, which is what licenses a causal conclusion. A confound is anything that varies alongside the independent variable, so that its effect cannot be separated from the effect of the manipulation. In a double-blind procedure neither the participant nor the experimenter knows which condition the participant is in. Keeping participants blind protects against placebo effects and participant demand, because their expectations no longer differ between conditions. Keeping experimenters blind protects against experimenter expectations, because they cannot treat the groups differently or score them with a bias, even unintentionally.

Question 7

Cherry found that separating two spoken messages was extremely difficult when both were recorded in the same voice on one tape, but easy when one message went to each ear. Explain what that contrast reveals about the basis on which attention selects, and how Broadbent's filter model accounted for it. What later finding forced the filter to be revised?

The contrast shows that selection is easy when the messages arrive on physically distinguishable channels and very hard when they share one. With the same voice on one tape, the listener struggled, closed his eyes and replayed phrases many times, and relied on knowledge of which words are likely to follow which; when the messages were built from clichés with arbitrary junctions, they could not be separated at all. With one message to each ear, the listener could follow either at will. So attention appears to select on simple physical properties such as ear, pitch and voice rather than on meaning. Broadbent's filter model captured this: incoming messages are held briefly in a sensory store, a filter selects one on its physical characteristics, and only that message passes on for analysis of meaning, so selection is early. The model had to be revised when Treisman showed that listeners follow a meaningful passage when it switches to the rejected ear, and Moray showed that people notice their own name there. Unattended messages are therefore attenuated, turned down like a volume control, rather than blocked.

Question 8

Why should a computer program that proved theorems in logic have mattered to psychologists who study people? Refer in your answer to the behaviourist objection to talking about internal mental states.

Behaviourists rejected talk of internal mental states such as goals, strategies or memory because these could not be observed and so seemed unscientific. The Logic Theorist was a physical system that manipulated symbols to prove theorems, working backwards from a goal by applying a small set of strategies and holding expressions in a limited working memory. It showed that internal states and processes, the very things behaviourism ruled out,

could be specified precisely, implemented and tested. If a machine could properly be described in terms of goals, strategies and working memory, there was no principled reason to refuse that vocabulary when describing a person. Its designers also described it with a diagram of labelled stages and arrows, the ancestor of the box-and-arrow models of cognitive psychology.

Question 9

Distinguish specificity, population, and sparse coding. Why is strict specificity coding an implausible account of how you recognise a particular familiar face, and what does sparse coding offer instead?

In specificity (labelled-line) coding, one neuron is tuned to one stimulus or quality, so its activity signals that quality, as with pain receptors. In population coding, each neuron responds to a range of values and the percept is carried by the pattern of firing across many neurons, as when colour is recovered from the combined responses of the three cone types. In sparse coding, a stimulus is represented by a small subset of neurons firing while the great majority stay silent. Strict specificity coding of a familiar face would need a dedicated cell for every recognisable person and object, and losing that cell would lose the concept, which is implausible. Sparse coding keeps the narrow selectivity that recordings in higher visual and medial temporal areas actually find, but spreads each representation over a small group of cells rather than one. Olshausen and Field argue that this increases associative-memory capacity, makes the structure of natural signals explicit, eases read-out at later stages and saves metabolic energy.

Question 10

Two brain regions show strongly correlated activity, but there is no direct axonal tract between them. Explain how this is possible using the distinction between structural and functional connectivity. Why does the existence of resting-state networks such as the default mode network complicate the assumption that cognition happens only when a task has been set?

Structural connectivity refers to the physical wiring, the axonal tracts that anatomically link regions. Functional connectivity is a statistical relationship: two regions are functionally connected if their activity rises and falls together over time. The two need not coincide, because two regions can be coordinated by communicating through an intermediate hub, so correlated activity is evidence of coordination rather than of a direct cable. Resting-state networks complicate the idea that cognition happens only when a task is set, because even when a person lies in a scanner doing nothing in particular, activity organises itself into reproducible networks. The default mode network is most active precisely when attention is not directed outward and is associated with mind-wandering and self-referential thought, so a great deal of brain activity is not in the service of any task the experimenter has imposed.

Question 11

A study reports that two regions are active while participants perform a task. Explain why this does not show that either region is necessary for the task, and describe how TMS could be used to test both necessity and timing. Give two limitations of that approach.

Imaging is correlational. Activity in a region during a task shows that the region is engaged at the time, but it may be responding to something incidental, or merely be along for the ride, so activation alone cannot show the region is necessary. TMS intervenes rather than observes: a brief pulse induces a current that temporarily disrupts a patch of cortex, a reversible virtual lesion in a healthy brain. If disrupting a region reliably impairs performance, the region was doing part of the work, as when occipital TMS disrupted Braille reading in early-blind readers but not in sighted controls. Because a single pulse acts only briefly, varying its timing relative to the stimulus also shows when a region contributes (causal chronometry); in a choice reaction-time task, pulses slowed responses at about 140 ms over anterior premotor cortex, 180 ms over posterior premotor cortex and from 220 ms over sensorimotor cortex. Limitations include coarse and uncertain spatial resolution, with effects possibly spreading to connected regions; access only to surface cortex, so structures such as the hippocampus and basal ganglia are out of reach; effects that can facilitate rather than disrupt; and a seizure risk with repetitive TMS, which requires safety screening and ethical approval.

Chapter 2: Attention

Question 1

In a shadowing task, what can people report about the unattended message, and what can't they? What does this tell you about the limits of processing?

People can shadow and report the attended message easily, but of the unattended message they can report only crude physical properties, such as whether the voice was male or female, or a change in pitch. They typically miss its meaning and even striking changes to it: a switch from English to German, or the same word repeated 35 times. This shows that the capacity to process meaning is tightly limited, so the system must select what it analyses in depth, and the question becomes where in the stream that selection happens.

Question 2

State the key problem with Broadbent's early filter, and explain how Treisman's attenuation model solves it.

Broadbent's filter selects one channel on physical features and blocks the rest completely before meaning is analysed. The problem is that meaning from the unattended channel sometimes gets through: people hear their own name in the ignored ear, they follow meaning across ears (reporting "Dear Aunt Jane" from "Dear 7 Jane" and "9 Aunt 6"), and in Treisman's own studies they followed a story when it switched to the wrong ear. Treisman kept selection early but replaced the all-or-nothing gate with an attenuator. The unattended message is turned down rather than switched off, and whatever survives reaches a dictionary unit in which each word has its own recognition threshold. Important or frequent words, above all your own name, have low thresholds and can be recognised even from a weakened signal. Selection is early, but leaky.

Question 3

In MacKay's experiment, participants could not report the word played to the unattended ear, yet that word shifted how they interpreted the attended sentence. Explain why this result is fatal to a strictly early filter, and what Treisman's attenuator would say about it.

The unattended word could not be reported, yet it biased the interpretation of the ambiguous attended sentence towards a riverbank or a financial bank. To do that, the word must have been processed all the way to its meaning. A strictly early filter blocks the unattended channel before any semantic analysis, so it predicts no such influence; the result is therefore fatal to it. Treisman's attenuator can accommodate it: the unattended word is weakened rather than blocked, and the attended sentence has primed the related meanings, lowering their thresholds in the dictionary unit, so the attenuated word can still activate its meaning and shape interpretation without reaching report. (MacKay's result is also exactly what Deutsch and Deutsch's late-selection model predicts.)

Question 4

Explain how load theory accounts for both early-selection-like and late-selection-like results with a single mechanism.

Load theory holds that perceptual capacity is limited and is spent involuntarily on whatever is present. When the relevant task carries high perceptual load (many items or a demanding discrimination), the target uses up all the capacity and nothing is left for distractors, so they are not processed and selection looks early. When the task carries low perceptual load, the spare capacity spills over onto the distractors whether or not you want it to, so they are processed and intrude, and selection looks late. A single mechanism, the amount of capacity left over, therefore produces both patterns, with early and late selection as two ends of a continuum set by load.

Question 5

The cross task varies load while holding the display identical. Why does that design matter, and what alternative explanation does it rule out?

In the cross task the display is identical in both conditions: a briefly shown cross with one green and one blue arm, one slightly longer. Under low load participants report which arm is green; under high load they report which arm is longer. Because the stimulation is the same and only the judgement differs, any difference in what else is noticed, such as an unexpected peripheral shape, cannot be due to differences in the display, for example more items, more clutter or different visual crowding. It isolates the demand placed on perceptual capacity as the cause.

Question 6

Macdonald and Lavie found that detection sensitivity (d') for a peripheral shape fell under high perceptual load with no change in response criterion. Explain why the criterion result is essential to their interpretation.

Detection sensitivity (d') measures how well a person can tell stimulus-present from stimulus-absent trials, whereas the response criterion measures how willing they are to say "present". If high load had simply made participants more conservative, more reluctant to report the shape, detection rates would fall through a criterion shift, and the result could be explained by a decision or reporting bias rather than by perception. Because d' fell while the criterion stayed the same, the drop reflects a genuine reduction in what was perceived. Combined with the design features that ruled out forgetting and neglect of the detection task, this is what licenses the conclusion that load reduces perception itself.

Question 7

Load reduces the V1 response to an image that observers never consciously see. What does that finding tell you about the relationship between attention and awareness?

Using continuous flash suppression, the tool images were invisible under both low and high load, yet their V1 response still shrank under high load (Bahrami et al., 2007). The effect of attention therefore cannot depend on awareness, because it operates on stimuli that never reached awareness in either condition. Attention is not simply a gate on consciousness; it is a limit on processing, much of which was never going to become conscious anyway. Attention and awareness are therefore dissociable.

Question 8

Load theory distinguishes perceptual load from cognitive (working-memory) load, and the two make opposite predictions about distraction. State both predictions, and explain why they are not a contradiction.

High perceptual load reduces distraction, because the relevant task consumes perceptual capacity and leaves none for distractors. High cognitive (working-memory) load increases distraction: when executive control processes are busy holding digits or tracking a changing sequence, they are less able to keep the task priorities straight, so irrelevant items win more often, as in the sandwich task results. The two are not contradictory because they tax different resources. Perceptual load acts on the capacity to perceive; cognitive load acts on the control that maintains what counts as relevant. Within working memory the distinction is finer still: maintenance of visual material behaves like perceptual load, while loading control increases awareness of distractors, especially when the distractor competes with the target.

Question 9

Summarise the dilution account of load effects and the reply to it. What kind of result would count as evidence against load theory, and why is that harder to specify than it looks?

The dilution account (Tsal & Benoni, 2010) argues that many load manipulations add neutral items to the display, which raises load but also dilutes the distractor's influence because it now competes with several items. When they kept the number of items constant but made the target pop out (low load, high dilution),

interference disappeared, so they attributed display-size effects to dilution rather than capacity. Lavie and Torralbo (2010) replied that spillover under low load still goes somewhere, in dilution displays to the neutral letters near the target, which then produced the response-competition effects, and that peripheral distractor effects return when spillover favours them. A common current position is a hybrid in which both effects reflect competition between stimuli for neural representation. Evidence against the theory would be distractor processing under genuinely high perceptual load. That is hard to specify because there is no measure of load independent of its effects: high load is defined as load sufficient to stop distractor processing, so interference under supposed high load can always be blamed on a weak manipulation. This circularity makes clean refutation difficult.

Question 10

Memory for a scene is better, not worse, when the scene coincides with a target in an unrelated detection task. Why is that awkward for the claim that perceptual capacity is spent obligatorily and completely?

Load theory claims that capacity is always fully used, so whatever the target consumes is taken from everything else. Detecting the white target square demands attention, so on that account the scene shown at the same moment should be remembered worse. In the attentional boost effect it is remembered better, by roughly 10 to 20 percentage points, and the obvious alternatives (distinctiveness, arousal, spatial cueing) have been ruled out. If capacity is always in use, a transient boost has nothing to draw on; if capacity expands whenever something relevant happens, it becomes unclear what would count as evidence against the theory. Swallow and Jiang propose a brief enhancement selective in time but not in space or modality, plausibly driven by phasic noradrenaline release from the locus coeruleus.

Question 11

Distinguish overt from covert attention and endogenous from exogenous orienting. What is inhibition of return?

Overt attention shifts by moving the eyes to a location; covert attention shifts without any eye movement, as when you watch someone out of the corner of your eye. Exogenous orienting is reflexive and stimulus-driven, triggered by a sudden peripheral event whether or not it is useful (the elephant reacting). Endogenous orienting is voluntary and goal-driven, triggered by a central symbolic cue such as an arrow that must be interpreted (the rider choosing). Inhibition of return is a signature of the reflexive system: an exogenous cue first speeds responses at its location, but from roughly 300 ms after the cue responses there become slower, as if attention is discouraged from returning to a place it has already visited.

Question 12

Distinguish location-based from object-based attention. What is the same-object advantage, and to which kind of orienting does attentional capture belong?

Location-based attention selects a region of space, as in the Posner cueing paradigm. Object-based attention selects a perceptual object, so that attending to part of an object spreads the benefit across the whole of it. The same-object advantage is the finding that a target is detected faster when it appears elsewhere on the attended object than at an equally distant point on a different object (Duncan, 1984; Egly et al., 1994). Attentional capture, in which a salient, sudden or distinctive stimulus seizes attention automatically whether or not it is relevant, belongs to exogenous orienting.

Question 13

What is an illusory conjunction, and how does it support the two-stage claim of Feature Integration Theory? Predict the search slopes for feature versus conjunction search.

An illusory conjunction is a wrong combination of correctly registered features, such as reporting a red triangle when the display held a red item and a triangle that were not one object. They occur when a display is flashed briefly and attention is loaded elsewhere, and disappear when observers have time to attend to each location. This supports Feature Integration Theory's two stages: features are registered separately and in parallel before

attention (so the colour and the shape are both detected), and focused attention is needed to bind them at a location (so without it they recombine at random). For search, a target defined by a single feature pops out, so search time stays roughly flat as set size increases. A conjunction target requires binding at each location in turn, so search time rises steeply, roughly linearly, with set size.

Question 14

Why does a hands-free phone conversation impair driving as much as a hand-held one? Frame your answer in terms of capacity and inattention blindness.

The main danger of phone use while driving is cognitive rather than manual. A conversation consumes the limited attentional capacity that noticing hazards requires, whether or not the driver's hands are free, and it does so far more than listening to the radio, because conversation demands active engagement. With capacity consumed, drivers show inattention blindness: they fail to register events they have looked straight at. Removing the handset does nothing to restore that capacity, so hands-free and hand-held calls impair driving to a similar degree. Only around two and a half per cent of people seem able to drive unimpaired while on a phone (Watson & Strayer, 2010).

Question 15

Under a consistent mapping, extensive practice turns search into an automatic process; under a varied mapping it does not. Explain why the mapping matters, and name one cost of a process having become automatic.

Under a consistent mapping a given item is always a target and never a distractor, so practice builds a stable association between the item and the response, and detection can become automatic: fast, effortless, largely unaffected by the number of distractors, and hard to switch off. Under a varied mapping today's target is tomorrow's distractor, so no stable association can form and search stays controlled, slow and serial. One cost of automaticity is that the process is hard to suppress when it is unwanted. Another is that attending to an over-learned skill can disrupt it, which is what happens when people choke under pressure.

Question 16

Give one example of attending to something you are not conscious of, and one of being conscious of something you are barely attending to. What does the pair show about the relationship between attention and awareness?

Attending without awareness: under continuous flash suppression, attentional load still reduces the V1 response to tool images that the observer never sees, and a masked number can guide a response without being consciously seen. Awareness with little attention: you can be aware of the general hum of a room, or that someone is standing at the edge of your vision, while your attention is on a conversation. The pair shows that attention and awareness are distinct. Attention is a limit on processing that can operate below awareness, while awareness depends on further conditions, plausibly reciprocal, synchronised exchange of signals across brain regions. We need attention for a great deal, but conscious awareness for less than we assume.

Chapter 3: Short-Term and Working Memory

Question 1

What did the whole-report versus partial-report difference in Sperling's experiment reveal about the capacity and decay of iconic memory? Why could the participants not have been preparing just one row in advance?

With whole report, people could name only four or five letters, which seemed to be all they had taken in. With partial report, a tone after the display cued one row, and people reported about three of its four letters. Since any row could be cued, about three-quarters of every row, and so most of the array, must have been available when the tone sounded. Iconic memory therefore has a large capacity. When the cue was delayed, partial-report performance fell to whole-report levels within about a second, showing very rapid decay: most of the array fades before it can be read out. Participants could not have prepared one row in advance because the cue came after the display and was unpredictable; preparing only one row would have produced good performance on a third of trials and poor performance on the rest, rather than consistently high accuracy on whichever row was cued.

Question 2

H.M. and K.F. had opposite memory deficits. Set out the double dissociation precisely, and explain why the pair is stronger evidence for separate stores than either case alone.

H.M. had intact short-term memory (he could hold a short string of digits normally) but could not form new long-term memories after his medial temporal lobe surgery. K.F. had the reverse: a digit span of one or two items, but broadly functional long-term learning and retention. Each was normal on exactly what the other could not do. A single case, such as H.M. alone, could be explained as long-term memory simply being more fragile or demanding, so that general damage would affect it first. The crossed pair rules out that explanation, because no single dimension of overall memory strength can produce both patterns. The two stores must depend on separable systems.

Question 3

In the Brown-Peterson task, why must participants count backwards during the delay? Explain how Keppel and Underwood's trial-order finding challenges a decay account of the result.

Counting backwards in threes occupies rehearsal, so participants cannot keep the trigram alive by repeating it; without that, the task would measure how long people can rehearse rather than how long information lasts unrehearsed. The Petersons attributed the steep forgetting to decay. Keppel and Underwood noticed that forgetting was minimal on the first trial and grew over later trials. If the cause were time-based decay, the first trial should look like any other, since the same time passes. What builds up across trials is similar material, so the better explanation is proactive interference: earlier trigrams compete with the current one at retrieval.

Question 4

Explain chunking and use it to interpret Chase and Simon's chess result. Why did the experts' advantage disappear for random boards, and what does that imply about the independence of the short-term and long-term stores?

A chunk is a meaningful package of information that occupies one unit of short-term capacity; using knowledge to recode several items into one chunk, as when eight digits become a familiar date, lets more raw information be held. Chess masters reconstructed far more pieces than novices from briefly shown positions taken from real games, because their long-term store of familiar configurations let them encode several pieces as a single chunk. On random boards there were no familiar patterns to recognise, so the advantage almost vanished (only a small residual advantage appears in larger samples). The masters did not have larger short-term memories. This

implies the short-term and long-term stores are not independent: what can be held in short-term memory depends on the contents of long-term memory.

Question 5

What does the Alvarez and Cavanagh result imply about whether short-term capacity is best measured in items or in information? Outline the slot and resource accounts and state how they differ.

Alvarez and Cavanagh found that the number of objects people could hold fell as complexity increased, from about four coloured squares to fewer than two shaded cubes, suggesting capacity is set partly by information per item rather than by a fixed count, with a ceiling of about four or five for the simplest items. Awh, Barton and Vogel argued this reflected harder change detection for similar complex objects, and when changes were easy to detect, the number held was about constant. On a slot account, visual working memory holds a small fixed number of items at fixed resolution: an item is either stored or absent, and complexity costs slots. On a resource account, a continuous pool of precision is shared among any number of items, so holding more items makes each less precise, with no hard item limit. The debate remains unresolved.

Question 6

Describe the evidence for the phonological loop from the phonological similarity effect, the word-length effect, and articulatory suppression. Why does the word-length effect suggest a time-based rather than an item-based capacity, and why does articulatory suppression strengthen the case?

Phonological similarity: lists of similar-sounding letters (b, d, g, p) are recalled worse than dissimilar ones even when read silently, so the store uses a sound-based code. Word length: lists of short words are recalled better than lists of long words with the same number of items. If capacity were a number of slots, length should not matter; that it does implies a time-based limit, about two seconds of speech, set by how much can be re-articulated before the earliest item fades. Articulatory suppression: repeating an irrelevant sound ties up the articulatory process and, for visual lists, abolishes both effects. This strengthens the case because it shows both effects depend specifically on the articulatory rehearsal process rather than on some general property of the words. With spoken lists, suppression removes the word-length effect but not the similarity effect, because heard speech reaches the sound store directly.

Question 7

What is the function of the central executive, and why was the episodic buffer added to the model? Refer to the chess-master result in your answer.

The central executive is the control system: it directs attention, inhibits irrelevant material, plans and sequences, updates the contents of the stores and integrates their outputs, often summarised as shifting, updating and inhibition. The original model had no way to bind the verbal and visual stores together or to explain how long-term knowledge enters and structures working memory. Yet we hold far more of a meaningful sentence or a real chess position than of a random list. Chess masters remember real positions so well because long-term knowledge of configurations is shaping what they hold. Baddeley (2000) added the episodic buffer, a limited store run by the executive that integrates information across the subsystems and with long-term memory, to accommodate exactly this.

Question 8

Loading working memory can either protect you from distraction or expose you to it. Explain how that can be true of the same system, using the maintenance-versus-control distinction.

Working memory has components that do different jobs. Loading maintenance, for example holding a set of coloured squares, occupies visual representational capacity and behaves like perceptual load: it reduces awareness of a peripheral distractor, protecting you from it. Loading control, as in the sandwich task with six digits or changing sequences, taxes the executive processes that keep task priorities straight, so irrelevant items win more often and distraction increases. The same system can therefore protect or expose you depending on which component is loaded, which is why "working memory load" has to be specified.

Question 9

How does Cowan's embedded-processes model differ from Baddeley's, and what kind of experimental result would favour one over the other?

Baddeley's model treats working memory as a set of dedicated stores (loop, sketchpad, buffer) coordinated by an executive. Cowan's embedded-processes model treats it as the currently activated portion of long-term memory, with a narrower focus of attention holding about four chunks; there are no separate stores. For Baddeley, interference between two tasks depends on whether they recruit the same subsystem, so a visual and a verbal task should coexist well. For Cowan, interference depends on the similarity of the material, regardless of modality. A result favouring Baddeley would be strong interference between dissimilar contents that share a subsystem and little between similar contents in different subsystems; the reverse pattern would favour Cowan. Cowan's account also handles expertise effects naturally, since activated long-term knowledge is what working memory is made of.

Question 10

What makes the oculomotor delayed-response task a clean test of working memory? Refer to the fixation requirement in your answer.

The monkey had to keep fixating a central spot throughout the delay, with nothing on the screen, and only after the fixation point went off could it look to where the cue had been. Because it could not move its eyes to the target during the delay, it could not solve the task by simply keeping its gaze, or a prepared eye position, on the cued location. Whatever bridged the gap had to be an internal representation of the location, so delay activity recorded during that interval reflects working memory.

Question 11

What is a memory field, and why did Funahashi and colleagues choose that name? Explain how the error-trial and delay-lengthening findings strengthen the claim that delay activity is the memory rather than an aftereffect of the cue.

A memory field is the region of visual space for which a prefrontal neuron fires during the delay: the neuron fires tonically when the remembered cue was in a particular direction, and weakly or with suppression for the opposite direction. Funahashi and colleagues named it by analogy with the receptive field of a visual neuron and the movement field of an oculomotor neuron: a receptive field describes the space a neuron responds to, a memory field the space it remembers. When the delay was lengthened from three to six seconds, the firing expanded to fill it, so the activity tracks the memory requirement rather than persisting for a fixed time after the cue. On error trials, delay activity was truncated or absent, so when the firing failed the memory failed. That correspondence rested on only nine neurons, so it should be held tentatively.

Question 12

Recording, lesion and clinical evidence are said to converge on the prefrontal cortex. Explain what each kind of evidence contributes that the others cannot, and why backward digit span is more informative here than forward digit span.

Recording shows what happens: prefrontal neurons hold a location during the delay, but this is correlational. Lesions show necessity: monkeys with prefrontal damage fail the delayed-response task, and unilateral damage produces errors for targets in the opposite visual field, matching the memory-field organisation. Clinical evidence shows the same pattern in humans: patients with prefrontal damage from stroke, tumour or aneurysm, and people with frontotemporal dementia, are impaired on demanding working-memory tasks. Backward digit span is more informative than forward span because it requires holding a sequence and manipulating it, not merely echoing it. Prefrontal patients often have normal forward span but impaired backward span, suggesting that what prefrontal damage costs is mainly the executive manipulation rather than storage, consistent with the sensory-recruitment view that content is stored posteriorly and prefrontal cortex supplies control.

Question 13

What is the contralateral delay activity, and why is it a more useful measure of the contents of working memory than a measure that tracked task difficulty?

The contralateral delay activity (CDA) is a sustained negative voltage over the hemisphere opposite to the side of a display whose items are being held in visual working memory. It persists through the retention interval and its amplitude increases with the number of items held, levelling off at each person's capacity. A measure that tracked task difficulty would rise for any demanding condition and could not distinguish holding more items from simply working harder. Because the CDA tracks the number of objects stored, it gives a read-out of what is actually in memory, which is what made it possible to see whether irrelevant items had been stored.

Question 14

Explain how Vogel and colleagues' findings connect working-memory capacity to attention, and why the claim that low-capacity individuals may store more information is not a contradiction. What did the add-versus-exclude experiment rule out?

Participants held two relevant items, four relevant items, or two relevant items with two irrelevant ones to ignore. High-capacity individuals showed a CDA like the two-item condition in the mixed display, so they excluded the irrelevant items; low-capacity individuals showed a CDA like the four-item condition, so they stored them too. Capacity and filtering efficiency were strongly correlated, linking working-memory capacity to an attentional skill. It is not a contradiction that low-capacity individuals may store more, because much of what they store is irrelevant: part of a large effective capacity is not admitting the wrong things. The add-versus-exclude experiment showed both groups added new items with perfect additivity, while only exclusion differed, ruling out the possibility that low-capacity individuals were simply poor at controlling working memory in general.

Question 15

In high-capacity participants, delay activity rose briefly towards the four-item level before falling back. Why does that time course complicate the description of filtering as a gate?

A gate would block irrelevant items at the door, so delay activity should stay flat at the two-item level. Instead the CDA in high-capacity participants briefly rose towards the four-item level when the irrelevant items appeared and then dropped back. The irrelevant items were admitted and then ejected. What distinguishes good filterers is a rapid correction after the junk is already inside, not a barrier that keeps it out. This shifts the question from where the selection point sits, as in the early-versus-late debate, to how quickly the system can undo its own admissions.

Question 16

Panichello and Buschman found that a classifier trained on selection from memory could decode attention to stimuli in prefrontal cortex but not in parietal cortex or V4. What does that dissociation suggest about where cognitive control is implemented? Give one respect in which selection and attention were nonetheless shown to differ.

In lateral prefrontal cortex, the code for selecting an item from memory generalised to the code for attending to a stimulus, and control information appeared there first before propagating backwards to parietal cortex and V4. In frontal eye fields, parietal cortex and V4 the two kinds of control were represented independently. This suggests prefrontal cortex acts as a domain-general controller issuing the same selection signal for memories and stimuli, while posterior regions keep separate codes. Selection and attention differed in competitive suppression: attention suppresses the representation of unattended neighbours, but selection from working memory did not reduce information about the unselected memory in prefrontal or parietal cortex. (Selection also re-formatted the chosen item into a shared subspace used to guide the response, and weaker alignment went with less accurate reports.)

Chapter 4: Long-Term Memory: Structure

Question 1

Why does a filled delay after a word list abolish the recency effect but leave the primacy effect intact? What does that selective pattern tell you that the shape of the curve alone does not?

A filled delay, such as thirty seconds of counting aloud, prevents rehearsal and empties the fragile short-term store, so the last few items, which were still being held there at the end of the list, are lost and recency disappears. The early items were rehearsed more and encoded into long-term memory, which the delay cannot reach, so primacy survives. The curve alone is only a shape that a single-store account could in principle produce. A manipulation that affects one end and not the other turns the two-store interpretation into evidence for two systems.

Question 2

Explain why long-term recency effects are awkward for the standard two-store reading of the serial position curve. Does the Glanzer and Cunitz result survive the objection?

Recency advantages also appear over intervals far too long for any short-term store: people recall the most recent rugby matches they attended over several seasons, or the most recent restaurants they visited over months, most readily. So a recency advantage on its own cannot prove that a short-term store is responsible, and some theorists argue both ends of the curve reflect one retrieval process working on different degrees of distinctiveness. The Glanzer and Cunitz result survives the objection, because the selective effect of the filled delay, abolishing recency while sparing primacy, still needs explaining. "Recency equals short-term memory" is a serviceable rule rather than a law.

Question 3

What is release from proactive interference, and why does it imply that short-term memory can code by meaning? Identify precisely what changed and what did not at the point of release.

Proactive interference is the build-up of old learning that impairs new learning. Over successive trials of three words from one category (fruits), each followed by a few seconds of counting backwards, recall falls as earlier items intrude. When the category switches (to professions), recall jumps back to about its original level: release from proactive interference. At the switch, everything physical stayed the same (number of words, presentation, test), and only the meaning of the items changed. The interfering system must therefore be organised by meaning, and because recall came only seconds after presentation, that semantic code is operating in short-term memory.

Question 4

Distinguish recall from recognition. Why should you be cautious about concluding from a recall failure that material has been forgotten?

In recall you must generate the material yourself, either with no support (free recall) or with a hint such as a category name (cued recall). In recognition the item is supplied and you only decide whether you have met it before, which requires a decision rather than a search. Performance differs greatly between them: from fifty categorised words people might produce about a third in free recall, but around three quarters when cued with the categories, even though nothing was added to memory in between. A recall failure therefore shows only that the material could not be retrieved under those conditions, not that it has gone. It may be available but not accessible.

Question 5

What are remember and know judgements, and how do they connect to Tulving's distinction between autonoetic and noetic awareness?

When people recognise an item, they can say whether they remember it, meaning they recollect the encounter with some of its context returning, or merely know it was there without any sense of the episode. The same correct recognition response can rest on either. Tulving distinguished autonoetic awareness, the self-knowing of episodic retrieval that amounts to mental time travel, from noetic awareness, knowing that something is so without reliving it. Remember judgements are the laboratory measure of autonoetic awareness, and know judgements of noetic awareness, which makes Tulving's distinction measurable in ordinary participants.

Question 6

Explain why H.M. and K.F. together provide stronger evidence than either alone. Define double dissociation in your answer, and state one thing a double dissociation does not establish.

A double dissociation is a crossed pair: damage in one case impairs ability A and spares B, while damage in another impairs B and spares A. H.M. had normal short-term memory but could not form new long-term explicit memories; K.F. had a digit span of one or two items but formed long-term memories more or less normally. If memory were one store damaged to different degrees, whoever was worse off would be worse at everything; no amount of damage to one store produces both profiles, so the crossing establishes at least partly separate machinery. A double dissociation does not establish that the systems are wholly independent, or that each maps onto one tidy brain region.

Question 7

What does Clive Wearing's case add to H.M.'s, given that the two profiles are so similar?

Clive Wearing reached a similar anterograde profile by a different cause, a brain infection rather than surgery, which is a converging case: it reduces the risk that something idiosyncratic about H.M., his surgery or history, explains the pattern. His case also extends the picture. His retrograde amnesia is far denser than H.M.'s, his semantic as well as episodic memory is impaired, and he cannot hold new experiences for more than about thirty seconds, yet he can still conduct and play the piano. That shows implicit procedural skill surviving even when explicit memory is very severely damaged.

Question 8

Distinguish episodic from semantic memory in terms of content, specificity, and the kind of awareness involved. Give an example of a dissociation between them.

Content: episodic memory holds personally experienced events, their what, where and when; semantic memory holds facts and concepts about the world. Specificity: an episodic memory is of one particular occasion; a semantic memory is general knowledge that holds regardless of when it was learned. Awareness: episodic retrieval is autonoetic, re-experiencing the past from the inside (mental time travel); semantic retrieval is noetic, knowing without reliving. Dissociation: K.C., after hippocampal and medial temporal damage, could not recall any personally experienced event but kept his semantic knowledge, while L.P., with lateral temporal damage, showed impaired semantic memory with relatively spared episodic memory.

Question 9

K.C. and L.P. show opposite profiles. Set out the dissociation and explain why the fMRI finding of overlapping retrieval networks is consistent with it rather than a problem for it.

K.C. lost episodic memory after hippocampal and medial temporal damage while semantic knowledge stayed largely intact; L.P., with lateral temporal cortex damage, lost semantic memory while episodic memory was relatively spared. Because each ability is lost in one patient and spared in the other, they form a double dissociation. fMRI in healthy people shows episodic retrieval leaning on hippocampus, medial temporal lobe and midline and parietal areas, and semantic retrieval on lateral temporal and frontal cortex, with substantial overlap. This is consistent with the patients because a double dissociation shows only that the systems are

separable, not that they are anatomically independent. Separable systems sharing some machinery fit the patient data better than a clean split would.

Question 10

What is semanticisation, and how does it change the character of an autobiographical memory over time? Which component is lost and which is kept?

Semanticisation is the gradual conversion of an episodic memory into semantic knowledge. An item often begins as an episode, the afternoon you learned that Canberra is the capital, and over time the episode dissolves and only the fact remains. Very old autobiographical memories therefore come to feel more like things known than events re-lived. What is lost is the autonoetic, re-experiencing component, the sense of being back in the scene; what is kept is the content.

Question 11

Why is episodic memory, rather than semantic memory, the one tied to a sense of self persisting through time? Use K.C. to explain what is lost and what survives, and distinguish personal semantic memory from episodic memory in your answer.

Semantic memory gives you a world: a store of what is the case, which would be much the same whoever had acquired it. Every episodic memory has you in it, at a particular time, which is why Tulving called its awareness self-knowing. After his injury K.C. could not recall any personal event and, asked about his personal future as well as his past, described his mind as blank: he could not place himself at any moment other than the present. Yet he knew a great deal about his own life as fact. That surviving knowledge is personal semantic memory, general facts whose subject happens to be you, such as your date of birth. What he lost was episodic memory, the ability to re-enter particular moments of his own past and project himself into the future.

Question 12

Give an example of priming affecting what someone perceives and an example of priming affecting what someone judges. Why is it significant that neither announces itself?

Perception: in repetition priming, a word or object seen recently is identified faster and from briefer or poorer input, so it is literally easier to see the second time. Judgement: in the propaganda effect, a repeated claim comes to feel more likely to be true, even when it is false. (Semantic priming, such as "nurse" speeding recognition of "doctor", or completing a word fragment as "library" after reading about borrowing books, also counts.) It matters that neither announces itself because the influence arrives as a change in how easy something is to process, not as a memory of the earlier exposure. People cannot discount an influence they do not notice, so implicit memory can shape what they see and believe without their knowledge.

Question 13

Name the three types of implicit memory and give an everyday example of each. How do H.M. and Clive Wearing show that implicit memory is separate from explicit memory rather than a weaker form of it?

Procedural memory, such as riding a bike or touch-typing; priming, such as completing a word fragment more readily after recently seeing a related word; and classical conditioning, such as feeling uneasy at a smell once associated with something unpleasant. H.M. learned new motor skills across days of practice while having no conscious memory of any practice session, and Clive Wearing still plays the piano and conducts despite dense amnesia. In each case the learning is clearly present in performance while recollection is absent. If implicit memory were only a weaker form of explicit memory, it should fail along with explicit memory; that it survives intact shows it depends on a separate system.

Question 14

Why can an expert often not report what they are doing while doing it? Explain why this follows from what procedural memory is, rather than being a failure of memory.

Procedural knowledge is knowing how, and it is often impossible to verbalise even in principle. As a skill becomes well practised it becomes automatic, and automatic performance does not lay down an explicit, episodic record of itself, so there is nothing reportable for the expert to retrieve, sometimes called expert-induced amnesia. The skill is perfectly retained and expressed in performance; it simply is not stored in a form that can be described. The everyday version is a familiar drive that went smoothly but leaves no recollection of the journey.

Question 15

The propaganda effect persists even when people are warned that the information may be false. Explain why, referring to the stage at which the effect operates, and say what this implies for attempts to correct misinformation by restating it.

The propaganda effect operates through familiarity rather than belief. Repetition makes a claim easier to process, ease of processing feels like recognition, and recognition feels like truth. The effect arrives as a feeling about the claim at the point of processing, not as a memory of having encountered it or a belief that can be reasoned with, so a warning that engages deliberate scepticism has nothing to act on. For correcting misinformation, this implies that restating a false claim in order to deny it may backfire, because the repetition increases its familiarity. A better strategy is to avoid repeating the false claim and to make the accurate version the familiar one.

Question 16

In what sense does the hippocampus consolidate rather than store memories, and why is that better treated as a slogan than as a precise division of labour?

The hippocampus and medial temporal lobe matter above all for forming new declarative memories, consolidating a fragile new experience into a durable trace that comes to be held, over time, in the cortex. This explains why medial temporal damage spares existing skills and much old knowledge while blocking new learning: the warehouse is intact, but the loading dock has gone. It is better treated as a slogan because medial temporal structures also contribute to the storage of declarative memories, so the division between consolidating and storing is one of emphasis, not a clean division of labour.

Question 17

What did Lashley's search for the engram find, and why is a null result about location a positive result about distribution?

Lashley trained rats on a complex maze, lesioned different cortical regions in different animals and retested them. He found that the size of the lesion predicted how much memory was lost, and its location did not. Failing to find a place where the memory was stored is informative: if no region is critical and the amount of tissue removed is what matters, the memory must be spread across many regions rather than filed in one. That is the classic demonstration that the engram is distributed, consistent with current evidence that memories are held across networks of cortical and subcortical areas.

Question 18

H.M.'s retrograde amnesia was temporally graded, which is what consolidation theory predicts. Explain why his medication and seizure history nevertheless make this weak evidence for the timescale of consolidation, and relate your answer to the cautions about single cases in §4.3.

Consolidation predicts that damage should spare distant memories more than recent ones, and H.M. lost memories from the years before surgery while childhood memories seemed normal. But he had frequent seizures for years before the operation and took a range of anti-epileptic drugs, both of which may have stopped those memories forming properly in the first place. If they were never laid down, their absence says nothing

about consolidation, so the gradient may reflect his pre-operative state as much as the loss of his hippocampus. This connects to §4.3: a double dissociation between two patients is strong evidence about architecture because the crossing does the work, but a quantitative claim drawn from inside one patient, such as the timescale of consolidation, is fragile because everything idiosyncratic about that person becomes part of the measurement.

Question 19

What are place cells and grid cells, and what does the Morris water maze show about what the hippocampus contributes to memory? Why is it significant that the deficit is specifically relational?

Place cells in the rodent hippocampus fire only when the animal is at a particular location, so the population traces out where the animal is. Grid cells in the entorhinal cortex fire at many locations forming a roughly hexagonal lattice across an environment; the overlap of grids with different spacing and offset indicates position. In the Morris water maze, intact rodents learn to use distant landmarks to swim straight to a hidden platform, whereas animals with hippocampal damage swim aimlessly despite normal vision and swimming. The hippocampus therefore contributes relational binding, linking the elements of an experience to their spatial and temporal context. That the deficit is specifically relational, rather than sensory or motor, matches what place and grid cells compute.

Question 20

Grid cells are often said to supply the coordinate system from which place fields are computed. Explain which part of that claim is well supported and which part the developmental evidence from rat pups calls into question. What role might border cells play?

Well supported: grid cells supply the hippocampus with a metric, a scale against which distance and direction can be measured, which is the Mosers' own claim. Called into question: the stronger idea that place fields are simply built by summing several grid patterns. If that were so, grid cells should mature before or with place cells. In rat pups leaving the nest at about two to two and a half weeks, sharp place fields are already present in many hippocampal cells, while grid cells are only weakly periodic and do not become strongly hexagonal until three to four weeks. Border cells, which fire near the geometric boundaries of an environment and are adult-like from a pup's first day outside the nest, may supply some of the input for early place fields. Alternatively, weak spatial input may be sharpened locally within the hippocampus.

Question 21

What are sharp-wave ripples, and what does disrupting them after training show? Explain why this evidence carries extra weight given the weakness of the temporal gradient in H.M.

Sharp-wave ripples are brief bursts of high-frequency hippocampal activity during rest and sleep, during which sequences of place cells that fired in earlier exploration are replayed, forwards or in reverse. Their frequency suits the induction of lasting synaptic change in CA1's targets, including distant cortex, which is what a mechanism for gradual cortical recruitment needs. Selectively disrupting ripples during rest after training impairs later memory for the task, establishing a causal link rather than a correlation. This carries extra weight because the classic human evidence, H.M.'s temporal gradient, is compromised by his seizures and medication. The consolidation account therefore rests on a mechanism that can be identified, blocked and shown to be necessary in animals, not only on that fragile case.

Question 22

How do concept cells bear on the choice between population coding, sparse coding and the grandmother cell from Chapter 1?

Concept cells in the human temporal lobe fire selectively for one specific person across photographs, the written name and the spoken name, so they code something close to the concept rather than a sensory pattern. They are too selective to be explained by broad population coding. But they are not grandmother cells in the strict sense, because many respond to several related concepts and each concept appears to be carried by many cells

rather than by one. That places them at the middle position, sparse coding, which Chapter 1 proposed as the plausible alternative to strict specificity coding.

Question 23

Distinguish habituation from sensitisation, and explain why both are classed as non-associative learning.

Habituation is the weakening of a response with repeated presentation of the same stimulus, as when a repeated poke in the back stops making you jump. Sensitisation is the strengthening of a response after a strong or significant stimulus, as when a loud bang makes you startle at a door you would normally ignore. Both are non-associative because neither involves learning a relationship between two things, as classical conditioning does; each is a change in responsiveness to a stimulus on its own.

Question 24

Free recall, cued recall and recognition of the same material give very different scores. What does that tell you about the difference between a memory being available and being accessible?

Free recall gives you no support, cued recall gives a hint, and recognition supplies the item itself, and scores rise markedly across the three for the same material. Since nothing is added to memory between the tests, the extra items were stored all along. A memory can be available (present in the store) without being accessible (retrievable under the current conditions). What usually limits performance is access, and the right cue can make an available memory reachable. Apparent forgetting is often a retrieval failure rather than loss.

Question 25

The T-maze result is described as a double dissociation established by lesion rather than by accident. Explain the difference and why it matters for the strength of the inference.

In a double dissociation by accident, the evidence comes from patients whose brain damage was caused by injury, disease or surgery, so the location and extent of the damage are not controlled, each case is a sample of one, and something idiosyncratic about a patient may do the work. In the T-maze work the experimenters deliberately lesioned the striatum in some animals and the hippocampus in others: striatal damage impaired habit learning (turning a fixed way for reward) while sparing hippocampus-dependent tasks, and hippocampal damage did the reverse. Because the lesions were chosen and could be applied across groups of animals, the inference that the declarative and procedural systems depend on different structures is much stronger, though the findings still need checking against humans.

Question 26

In what way is the amygdala's contribution to memory different in kind from the hippocampus's and the striatum's?

The hippocampus forms new explicit memories, and the striatum supports procedural habits; each is part of the system that forms and holds a particular kind of memory. The amygdala's contribution to emotionally arousing declarative memories is modulatory: it does not appear to store them so much as tune how strongly they are consolidated elsewhere. Amygdala damage in humans removes the normal memory advantage for emotional material rather than abolishing declarative memory. For conditioned fear the amygdala may hold part of the trace itself, since destroying the lateral amygdala neurons recruited during learning can erase an already-learned fear in mice.

Chapter 5: Long-Term Memory: Encoding, Retrieval and Consolidation

Question 1

A student cannot recall a term they revised last week. Set out the three stages at which the failure could have occurred, and say what evidence would distinguish them.

The failure could be at encoding: the term was never properly learned, for example because it was only read over or rehearsed by rote. It could be at storage: the trace was formed but has degraded or been disrupted, for instance by interference from similar material learned since. Or it could be at retrieval: the trace is intact but inaccessible because the right cue is missing. Evidence that would distinguish them: if a good cue (the first letter, the context in which it was learned, a recognition test) brings the term back, it was stored and the problem was retrieval. If recognition also fails, and the student studied only shallowly, encoding is the likely culprit. If the term was clearly learned (it was recalled on an earlier self-test) and is now lost even to strong cues, a storage problem is more likely. Only a retrieval failure is reversible on the spot.

Question 2

Distinguish maintenance from elaborative rehearsal, and explain why the modal model's treatment of rehearsal was incomplete.

Maintenance rehearsal is rote repetition that holds information in mind without engaging its meaning, such as repeating a phone number until you dial it; it keeps the item available briefly but builds little durable trace. Elaborative rehearsal processes meaning and connects the item to existing knowledge, and is far more effective at getting information into long-term memory. The modal model treated rehearsal as a single pump moving information from short-term to long-term memory, implying that more rehearsal means better storage. Because the same time spent in different kinds of rehearsal produces very different results, what matters is the kind of processing, not just its amount.

Question 3

Why does it matter that Craik and Tulving's participants did not know their memory would be tested? What alternative explanation does that design rule out?

Because the test was unexpected, participants had no reason to try to learn the words, so the depth of processing was set entirely by the orienting question (structural, phonemic or semantic), not by any intention to memorise. This rules out the explanation that semantic questions produced better memory because people tried harder or used deliberate memorisation strategies on those items. The memory difference can be attributed to the kind of processing itself.

Question 4

The levels-of-processing account has been accused of circularity. Explain the charge precisely, and describe how transfer-appropriate processing meets it. Use Morris and colleagues (1977) in your answer.

The charge is that depth is defined by its effect: a process is classed as deep because it produced good memory, and memory is said to be good because the process was deep. Without an independent measure of depth (processing time and effort do not track memory reliably), the theory risks re-describing the result rather than explaining it. Transfer-appropriate processing meets this by making a directional prediction in advance: memory is best when the type of processing at encoding matches that required at test, and this can fail. Morris and colleagues had participants encode words for meaning or for rhyme, then gave either a standard recognition test or a rhyming recognition test. Semantic encoding won on the standard test, but rhyme encoding won on the

rhyming test, so the lines crossed. Deep processing was not universally superior; it won when the test demanded meaning.

Question 5

Explain why the self-reference effect produces such good encoding, and connect your answer to what §4.4 said about episodic memory and the self.

Judging whether a word describes you produces better recall than ordinary semantic processing. The self is the most richly elaborated knowledge structure anyone has, so relating material to it creates a large number of connections and makes the trace distinctive, which are the two ingredients of deep processing. This connects to §4.4, where episodic memory turned out to be bound up with the self: every episodic memory has you in it at a particular time. Encoding material in relation to yourself attaches it to that same well-organised structure, so the self acts as a powerful encoding device as well as the subject of autobiographical memory.

Question 6

In Karpicke and Roediger (2008), Groups 2 and 3 both reduced their effort. Why did only one of them suffer at the one-week test? What does this tell you about how to revise, and why does it feel wrong?

Group 2 stopped restudying pairs once they had been recalled but kept testing all pairs; Group 3 kept restudying all pairs but stopped testing pairs once they had been recalled. A week later Group 2 recalled about 81 per cent, the same as the standard group, while Group 3 fell to about 36 per cent. Repeated retrieval was doing the work, and repeated study of material already learned added almost nothing. So revision should centre on testing yourself repeatedly, including on items you can already recall, rather than rereading. It feels wrong because rereading feels fluent and easy, and fluency is mistaken for learning, whereas retrieval is effortful and exposes what you cannot yet produce, which feels like failing even though it builds durable memory.

Question 7

State the encoding specificity principle and explain, using the divers, why it reframes forgetting as a retrieval problem. Why is the effect weaker for recognition than for free recall?

The encoding specificity principle (Tulving & Thomson, 1973) states that a retrieval cue is effective to the extent that it overlaps with what was encoded at learning: a cue works by reinstating the original encoding conditions. Godden and Baddeley's divers learned words on land or underwater and recalled better when the test environment matched the learning environment, so the external context had become part of the trace. On this view, much of what we call forgetting is not loss of the trace but a mismatch between the cues available now and those bound in at encoding. The effect is weaker for recognition because the test supplies the item itself, which is already a very strong cue and has less need of context. (A direct replication with sixteen divers did not find the same-context advantage, so the study is best taken as an illustration of the principle.)

Question 8

Why are self-generated retrieval cues more effective than cues generated by someone else, even when the other person's cues are perfectly sensible? Explain what Mäntylä's no-study condition was for and what it rules out, and say what practical advice follows about using a classmate's notes.

A cue works by overlapping with the particular way an item was encoded. Cues you generated yourself are guaranteed to match your own encoding, whereas another person's cues, however sensible, reflect their encoding, not yours. Mäntylä found 91 per cent recall of 504 words with self-generated properties as cues, but 55 per cent with another participant's properties. The no-study condition gave the same cues to people who had never seen the words, and they produced the target only 17 per cent of the time. This rules out the explanation that good properties simply describe a word well enough for anyone to reconstruct it: the cues work only in combination with the person's own encoding. Practically, a classmate's notes are someone else's retrieval cues and, on these numbers, worth roughly 60 per cent as much as your own, so it is better to make your own notes and questions, or at least to rework theirs into your own words.

Question 9

Distinguish retroactive from proactive interference, and explain why the two designs are so easily confused. Which memory suffers in each case?

In retroactive interference, new learning disrupts retrieval of older material: after learning list B, recall of list A suffers, so the old memory is the casualty (a new phone number blocking the old one). In proactive interference, old learning disrupts retrieval of newer material: having learned list A, recall of list B suffers, so the new memory is the casualty (producing last year's password). The designs are easily confused because both involve learning two similar lists and comparing with a control group that learns only one; the only difference is which list is tested, the earlier one (retroactive) or the later one (proactive).

Question 10

Why is decay so difficult to test as an account of forgetting? Explain how cue overload might unify interference with encoding specificity, and what would be lost if it did.

To show decay you would need an interval in which time passed but nothing else happened, with no interfering material. In a waking person time never passes empty, so any forgetting over time can be attributed to interference instead, which is the problem Keppel and Underwood raised for the Brown-Peterson task. Cue overload is the idea that a cue linked to many items points only weakly to each. On this view interference is what happens when the cue-to-item mapping becomes crowded with similar items, so it is a consequence of encoding specificity rather than a separate mechanism. This explains why similarity drives interference and why switching category releases proactive interference. What would be lost is interference as a distinct explanatory mechanism: the two literatures would be describing one phenomenon, and the question of whether interference adds anything beyond cue competition would disappear.

Question 11

Long-term potentiation is said to have the properties a memory mechanism ought to have. Name four such properties and say, for each, why a memory system would require it.

It is rapid, which a memory system needs because memories can be formed in a single episode. It is long-lasting, because memories persist. It is input-specific, so strengthening one pathway does not indiscriminately strengthen neighbouring ones, which a memory system needs if different memories are to be kept distinct. And it is associative, so that inputs active together are strengthened together, which is what binding the elements of an experience requires. In the hippocampus, NMDA receptors act as coincidence detectors, opening only when pre- and post-synaptic activity coincide, implementing Hebb's principle.

Question 12

A protein-synthesis inhibitor given after initial learning blocks a new memory but not an old one. In Nader and colleagues (2000) the same drug abolished an old, consolidated memory. What had to be done first, and why does the drug-alone and reminder-alone comparison matter to the argument?

Nader and colleagues first had to reactivate the consolidated fear memory by briefly playing the tone the day after conditioning, and only then infuse the protein-synthesis inhibitor into the amygdala. The memory was abolished. The comparison matters because the drug alone does nothing to an old memory (consistent with protein-synthesis inhibitors blocking only new consolidation) and the reminder alone does nothing lasting. Only the combination destroyed the memory. The most economical explanation is that reactivation returned the memory to a labile, protein-synthesis-dependent state, so it had to be re-stored, and blocking synthesis during that window prevented re-storage. Retrieval reopened the memory rather than simply reading it out.

Question 13

Contrast the standard and multiple-trace models of systems consolidation. What pattern of retrograde amnesia would count as evidence for each, and why does the doubt over H.M.'s temporal gradient matter here?

On the standard model the hippocampus rapidly binds the distributed cortical fragments of an experience, and through repeated reactivation cortico-cortical connections strengthen until the cortex supports the memory alone, so hippocampal dependence is temporary. On the multiple-trace model the hippocampus never fully relinquishes detailed episodic memories: each retrieval lays down a new trace, so vivid episodic memories stay hippocampus-dependent however old they are, and only semantic gist becomes independent. The standard model predicts a temporal gradient, with remote memories spared after hippocampal damage. The multiple-trace model predicts that episodic detail is lost across the whole lifespan while semantic gist shows a gradient. The doubt over H.M. matters because his gradient is the standard model's headline human evidence, and years of seizures and medication before surgery may have prevented recent memories forming at all, so the case carries less weight than usually assumed.

Question 14

Sleep is said to support consolidation both passively and actively. Explain both routes, and say why targeted memory reactivation is stronger evidence for the active one than replay observations alone.

Passively, sleep protects a fragile trace by removing the interference that waking experience would otherwise impose. Actively, during slow-wave sleep the hippocampus replays recent learning in sharp-wave ripples, in effect training the cortex. Observing replay alone shows only a correlation between replay and later memory. In targeted memory reactivation, a cue (an odour or a sound) is paired with material at learning and re-presented during sleep; cued material is remembered better, and Rudoy and colleagues found the benefit for the specific cued object locations. Because the experimenter controls which memories are reactivated and the benefit follows that choice, this turns the active account from an inference into a manipulation. A meta-analysis of about ninety studies finds a small-to-moderate benefit, clearest when cues are given in non-REM sleep.

Question 15

The crossover in Morris and colleagues (1977) appeared only for congruent acquisition items. Explain what congruency means in that design, and discuss how far this qualification weakens the case for transfer-appropriate processing.

Each word arrived in a frame it either fitted or did not: a sentence it completed sensibly or did not, or a word it rhymed with or did not. Congruent items are those where the answer was yes, so the word was integrated with its frame at encoding; incongruent items were merely judged against it. The crossover, rhyme encoding beating semantic encoding on the rhyming test, appeared only for congruent items. For incongruent items semantic encoding won on both tests. This narrows the case rather than refuting it: the most cited demonstration holds only when the item was integrated with its frame, and Morris and colleagues said the role of congruency needed clarification. The principle survives, but the famous crossover is conditional, and it is unwise to rest a strong argument on it alone.

Question 16

Explain how reconsolidation could account for the finding that flashbulb memories drift while remaining vividly confident, and say what evidence would be needed to show that it does.

Flashbulb memories feel photographic and are held with high confidence, yet they drift and distort over time. On a reconsolidation account, every time you retell or recall the event you reactivate the memory and return it to a labile state, and what is re-stored is the version you have just told, including any changes. Repeated retellings would therefore accumulate drift, while the vividness and rehearsal keep confidence high. To show that this actually happens, you would need evidence that recalling the memory opens it to modification in humans: for example, that the amount of drift tracks the number of retellings, that new information introduced around a retelling is absorbed into the memory (as in Hupbach and colleagues' reminder effect), or that blocking re-

storage after reactivation alters the memory. The account is plausible but has not been directly tested for flashbulb memories.

Question 17

Set out the four routes by which a reactivated memory might be modified, and name one practical and one ethical obstacle to using any of them clinically.

Sandrini and colleagues (2018) describe four routes. Behavioural means, such as interference or extinction, supply new content during the reconsolidation window, close to what exposure therapy does. Stressor events alter the hormonal conditions under which re-storage happens. Pharmacological agents act on the biology of re-storage, for example propranolol given after reactivation to blunt the noradrenergic arousal that would strengthen the memory. Non-invasive brain stimulation (tDCS or TMS) targets the regions involved in reactivation. A practical obstacle: reactivation does not reliably open a reconsolidation window, since it depends on timing, how strongly the memory is reactivated and how old and strong it is, and brain-stimulation effects vary considerably between individuals. Reactivating a traumatic memory is also itself distressing. An ethical obstacle: a memory is a record and sometimes evidence as well as a burden, so deciding when a memory should be weakened, and who decides, is not something the science can settle.

Question 18

Design a two-week revision plan for this unit that uses at least five principles from this chapter. Justify each choice by naming the principle it exploits, and say which of your choices will feel least productive while working best.

One model plan. Days 1 to 3: for each chapter, write your own questions and summary cards in your own words, with properties or cues you generate yourself (generation effect; self-generated cues, Mäntylä). Organise each chapter's material under its main headings and link examples to your own experience (organisation; self-reference; elaboration). From day 2 onward, start each session by answering yesterday's questions from memory before looking at notes, and keep testing items even after you get them right (testing effect; Karpicke and Roediger). Spread each chapter across several sessions separated by days instead of finishing one chapter per day, interleaving chapters within a session (spacing effect). Write practice answers in the format you will actually be asked to produce, timed and closed-book (transfer-appropriate processing; encoding specificity, by studying in conditions like the test). Make vivid interactive images for lists of terms or studies (imagery). Sleep normally throughout and avoid an all-nighter before the test (sleep-dependent consolidation). The choices that feel least productive while working best are retrieval practice on items you already know and spacing, because both feel effortful and slow compared with fluent rereading, yet they produce the most durable learning.

Chapter 6: Everyday Memory and Memory Errors

Question 1

What makes autobiographical memory multidimensional, and why is the own-photographs versus laboratory-photographs comparison a better test of self-relevance than simply comparing vivid with dull memories?

Recalling a personal event recruits a broad network of brain regions rather than a single store, because the original experience bundled together vision, sound, emotion, place and self, and retrieval has to reassemble that bundle. Comparing vivid with dull memories confounds self-relevance with many other differences, such as emotional intensity, distinctiveness, rehearsal and content. In Cabeza and colleagues' study, people's own photographs and someone else's photographs showed the same locations, so what was depicted was held constant and the main difference was whether the memory was the person's own. Their own photographs produced stronger medial prefrontal and hippocampal activity, as well as stronger visual and parahippocampal activity, isolating the effect of self-relevance.

Question 2

Why does the loss of autobiographical memory in dementia amount to more than the loss of information? Refer to §4.4 in your answer.

Section 4.4 argued that episodic memory is what places a person at particular moments in their own past, which is why Tulving called its awareness self-knowing: the self that persists through time depends on the system that remembers past events. Autobiographical memory supplies the narrative a person uses to know who they are. In dementia, particularly Alzheimer's disease, its progressive loss therefore strips away not just stored facts but the scaffolding of identity, the sense of being a continuous someone with a history.

Question 3

The reminiscence bump shifts to a later age in people who emigrate in adulthood. Which of the three hypotheses does this support, and why is a prediction that could have failed better evidence than one that could not?

It supports the cognitive hypothesis, which says encoding is best during periods of rapid change followed by stability, when experiences are novel and distinctive. In people who emigrate in adulthood, the bump moves to the age at immigration, when the flood of novelty occurred, although the cultural life script did not change and identity formation had largely run its course. The hypothesis made a risky prediction: if the bump had stayed in adolescence and early adulthood, the hypothesis would have been in trouble. A prediction that could have failed and did not is stronger evidence than one compatible with any outcome, because only the former genuinely tests the theory.

Question 4

Why might the reminiscence bump be overdetermined? Explain how all three hypotheses could be partly correct at once.

Overdetermined means that several causes each push towards the same outcome, so the effect would appear even if one cause were removed. Adolescence and early adulthood are periods of rapid change and novelty (cognitive hypothesis), the period when a stable self-image forms, so self-defining events are encoded well and rehearsed (self-image hypothesis), and the period containing most culturally scripted milestones such as leaving home and a first job, which give retrieval a ready-made structure (life-script hypothesis). The hypotheses are not mutually exclusive, and each has distinctive support: the bump shifts with emigration, as the cognitive account predicts, but it appears for happy and important memories and not for sad or traumatic ones, as the life-script account predicts. All three can be partly correct at once.

Question 5

A friend insists their memory of hearing shocking news is burned in and therefore accurate. Using Talarico and Rubin (2003), explain why their confidence is no guarantee of accuracy, and say why the everyday-memory comparison condition was essential to the argument.

Talarico and Rubin recorded students' memories of hearing about the September 11 attacks the day after the event, together with an ordinary event from the same period, and retested each person once, 1, 6 or 32 weeks later. Consistency with the original report declined at much the same rate for the flashbulb memory as for the everyday memory. Confidence, however, stayed high for the flashbulb memory while it fell for the everyday one. So the friend's certainty reflects how vivid the memory feels, not how accurate it is. The everyday-memory condition was essential because it gave a matched comparison: both memories were recorded at the same time and tested on the same schedule, so the flashbulb memory could be compared against an ordinary memory rather than against intuition, revealing that only confidence differed.

Question 6

The amygdala stamps in emotional memories and each retelling rehearses them. Explain why these two mechanisms do not make flashbulb memories more accurate, referring to reconsolidation in §5.8.

The amygdala modulates how strongly emotional memories are consolidated, stamping in the surrounding detail, and narrative rehearsal consolidates the memory further with each retelling. Both make the memory stronger and more vivid, but neither checks that what is being stamped in or rehearsed is accurate. Section 5.8 adds that each retrieval may reopen the memory and return it to a labile state, so that what is re-stored is the version just told, including any distortions. Rehearsal could therefore make the memory more confident and more consistent with the latest retelling while carrying it further from the original event. This is a plausible mechanism rather than a demonstrated one, but it fits Talarico and Rubin's results.

Question 7

What is a schema, and what did Bartlett's War of the Ghosts demonstrate about how schemas act on memory over repeated retellings?

A schema is an organised packet of knowledge about how some part of the world usually works; a script is a schema for a sequence of events, such as eating at a restaurant. Bartlett had British participants repeatedly recall an unfamiliar Native American folk tale. With each retelling the story drifted towards their own cultural expectations: canoes became boats, and unfamiliar supernatural elements were dropped or rationalised into something more familiar. Schemas therefore act on memory over time, filling gaps with what was expected rather than what was experienced, so people remember the story their schema leads them to expect. The same structure that makes encoding efficient also quietly reshapes memory.

Question 8

Studying a list of related words often produces confident recognition of a related word that was never presented. Explain why this is a by-product of successful encoding rather than a failure of attention.

When you study a list of related words, you extract its gist, its shared meaning, rather than storing it as a string of unrelated sounds. A closely related word that was never presented fits that gist so well that it is generated from it, and the resulting sense of having seen it is indistinguishable from the sense of having seen the real items. The false memory is produced by exactly the meaning-extracting machinery that makes encoding successful. It is not a lapse of attention or a guess under pressure.

Question 9

Distinguish a pragmatic inference from a source-monitoring error, and give an example of each.

A pragmatic inference is going beyond what was said to what was implied, then remembering the inference as if it had been stated. Told that the karate champion hit the cinder block, people later recall that he broke it; told that the baby stayed awake all night, they may recall that it cried all night. A source-monitoring error is misattributing where a memory came from while its content is intact. Examples are judging a non-famous name

famous a day after reading it in a list, or remembering a claim but forgetting that it came from an unreliable website. The first adds content that was never presented; the second keeps the content and loses its origin.

Question 10

Explain how a source-monitoring failure underlies both the becoming-famous-overnight result and the misidentification of an innocent bystander in a lineup. What is preserved in each case, and what is lost?

In the becoming-famous-overnight study, reading a non-famous name gave it familiarity that survived a day's delay, while memory for the source ("I saw it in yesterday's list") faded. Left with unexplained familiarity, participants attributed it to the most plausible source: the name must be famous. In a lineup, an innocent bystander who was present at the scene has a familiar face, and if the witness has lost the source of that familiarity (seeing him reading to students, for example), it can be misattributed to his having been the perpetrator. In both cases familiarity is preserved and its label, where the familiarity came from, is lost. Source monitoring fails not by losing the memory but by losing its label.

Question 11

Why does the illusory-truth effect mean that a false claim need not defeat your reasoning in order to influence you? Relate your answer to fluency and to the Rider and the Elephant.

Repeating a claim makes it easier to process, and that fluency is misread as a sign of truth. The effect works through how the claim feels rather than through argument, so a false claim does not have to win a debate with your reasoning; it only has to become familiar. In the book's terms, the Elephant generates the feeling of truth automatically and the Rider receives it as if it were its own judgement, with nothing to inspect or rebut. That is why the effect survives an explicit warning that the material may be false.

Question 12

Lindsay (1990) found the misinformation effect was largest when the misleading source was hard to distinguish from the original. Why does that result place the misinformation effect inside source monitoring rather than alongside it?

Lindsay varied how easily the misleading information could be distinguished from the original event, in timing and presentation conditions, and told participants that nothing in the misleading narrative was correct, so any suggested detail they reported must have been misattributed to the event. When the sources were hard to tell apart, the suggested detail was reported on 27 per cent of misled items against 9 per cent of controls; when the misleading information came two days later under clearly different conditions, the difference nearly vanished (13 against 10 per cent), even though that group had heard the misinformation most recently. If misinformation simply overwrote memory, the most recent exposure should have done the most damage. Instead the damage depended on whether people could tell where a detail came from, which shows the misinformation effect is a source-monitoring failure, not a separate phenomenon alongside it.

Question 13

Ross and colleagues found 60 per cent misidentification of the bystander among witnesses who had seen him, and 18 per cent among those who had not. Explain why the comparison, rather than the 60 per cent alone, is what shows familiarity is doing the work.

Sixty per cent on its own could reflect many things: the bystander might simply resemble the culprit, the lineup might be unfair, or witnesses might pick someone whenever the culprit is absent. The control group faced the same lineup, with the same faces and the true robber absent, but had never seen the innocent man. Their 18 per cent rate shows how often he would be chosen for those other reasons. The only difference between the groups was prior exposure to his face in an innocent setting, and that more than tripled the rate. The comparison isolates familiarity as the cause.

Question 14

A lineup is run by an officer who knows which person is the suspect, with no warning that the culprit may be absent. Identify two things wrong with this procedure and the specific error each invites.

First, the administrator knows who the suspect is. Even unintentionally, they can steer the witness through cues and, after an identification, give confirming feedback, which inflates the witness's confidence and their reports of how good their view was (the post-identification feedback effect). A blind or double-blind administrator prevents this. Second, the witness is not warned that the culprit may be absent. Without that warning, witnesses treat the lineup as multiple choice and pick whoever best matches their memory, so when the culprit is absent an innocent person, especially a familiar one, is likely to be chosen.

Question 15

Why is a witness's confidence in court a poor guide to accuracy, and what would recording confidence at the moment of identification achieve?

Confidence expressed in court is measured long after the identification and can have been inflated in the meantime, especially by confirming feedback such as "Good, you identified the suspect", which raises confidence without changing accuracy (Wells & Bradfield, 1998). Repeated questioning and rehearsal can raise it further. So a witness may be confident precisely because someone told them they were right. Recording confidence at the moment of identification, before anyone reacts, captures a judgement uncontaminated by later feedback. It provides a baseline against which later expressions of certainty can be compared, and a discrepancy between initial and courtroom confidence is itself informative.

Question 16

Kaesler and colleagues (2020) found sequential lineups change response bias but not discriminability. Explain the distinction, why a reduction in false identifications is not by itself evidence of improvement, and why this matters for policy.

In signal-detection terms, discriminability is the witness's actual ability to tell a guilty face from an innocent one; response bias is how willing they are to choose anyone at all. Kaesler and colleagues found that sequential lineups make witnesses more conservative about choosing but leave discriminability little changed, and other analyses find a modest discriminability advantage for simultaneous lineups. A procedure that makes witnesses choose less often reduces false identifications, but it also reduces correct identifications, because both follow from the same reluctance to commit, so fewer false identifications alone is not an improvement. Only an increase in discriminability means witnesses are distinguishing better. This matters for policy because the sequential reform was adopted in many jurisdictions on the basis of reduced false identifications, a result that reflected a shift in bias rather than better memory.

Question 17

Set out the stages an intention passes through, and say which stage typically fails for a time-based intention and why.

An intention is encoded together with the action it requires; held across a retention interval while you do other things; then, in the performance interval, you must notice that the right moment has arrived; you initiate and execute the action; and finally you evaluate whether it went to plan. For a time-based intention, the stage that typically fails is noticing the right moment in the performance interval. Nothing in the environment announces that it is 8 pm, so you must monitor the time yourself while busy with other things, and self-initiated monitoring is easily lost.

Question 18

Convert the time-based intention "phone the clinic at 3 pm" into an event-based one, and explain in terms of cues why the change should help.

Make the call triggered by an event you know will occur around 3 pm: for example, "when I get back to my desk after the 2.30 lecture, I will phone the clinic", ideally with the clinic's number left on the keyboard, or set an

alarm for 3 pm. An event-based intention is cued by something in the world, so the environment does part of the remembering: the sight of the desk, or the alarm, retrieves the intention. This removes the need for self-initiated monitoring of the clock, which is the stage at which time-based intentions usually fail.

Question 19

Explain how algorithmic content selection would be expected to interact with the illusory-truth effect, and why that interaction offers an account of political polarisation that does not require anyone to be irrational.

An engagement-optimising algorithm shows you more of what you already respond to, so the claims you see repeatedly are not random but those you were already disposed to accept. Each repetition increases their fluency and therefore their felt truth through the illusory-truth effect, while source monitoring lets you remember the claim long after forgetting it came from an unreliable feed. If different communities are fed different repeated claims, each develops its own set of things that feel obviously true. Disagreement between groups then cannot be resolved by argument, because the beliefs were never established by argument in the first place. Polarisation can arise this way through ordinary memory mechanisms working as designed, without anyone being stupid, dishonest or irrational.

Chapter 7: Conceptual Knowledge

Question 1

Explain what is meant by calling a concept a pointer to knowledge, using an example of an object you have never encountered.

A concept is a mental representation of a category, and knowing that something belongs to it gives access to a wealth of general knowledge that did not come from the object itself. If you are told that an unfamiliar vehicle is a truck, you immediately know it has wheels and an engine, carries loads, runs on fuel and behaves in particular ways in traffic, even though you have never seen that truck. The concept points to stored knowledge that can be applied to the new instance, which is what lets us generalise from a few instances to an effectively unlimited number of new ones, while still noticing what is distinctive about a particular one.

Question 2

Why does the definitional approach succeed for square but fail for bird? What does family resemblance offer instead, and what does it cost us in precision?

The definitional approach requires features that are individually necessary and jointly sufficient. Square has such a definition (a four-sided polygon with equal sides and right angles), so anything meeting it is a square and nothing else is. For bird, any plausible definition fails: if birds fly, the penguin is excluded although it is clearly a bird. Necessary features can often be found, but a set that is also sufficient usually cannot. Family resemblance (Wittgenstein; Rosch and Mervis) holds that members share overlapping features without any one feature being common to all, so membership is graded rather than all-or-none. The cost is precision: there is no sharp boundary or test that settles membership, so borderline cases are genuinely uncertain.

Question 3

Rosch and Mervis gave a two-part account of typicality. State both parts, and use them to explain why a penguin is a poor bird in a way that mentioning flight alone does not capture.

An item is typical to the extent that (1) it has features that are frequent in its own category and (2) it lacks features that are frequent in other categories. The robin has many common bird features and few features belonging elsewhere, so it is highly typical. The penguin is a poor bird not only because it lacks a common bird feature, flight, but because it has features frequent in other categories: it swims and has fin-like wings, which belong more to fish and marine animals. Mentioning flight captures only the first half of the account; typicality depends on contrast with other categories as well as resemblance within one.

Question 4

Describe Rosch's priming study and explain precisely what it rules out. Why could a prototype that was merely a stored description not produce this result?

Participants heard a colour name such as "green" and then judged whether two colour patches were the same or different. On some trials they were faster when the colours were good, prototypical greens than when they were poor off-greens. This rules out the view that a prototype is only a description that can be produced on request: if it were, hearing the word would not change how quickly two colour patches could be compared. The category name must have activated a representation that was already biasing perception before the colours appeared, so the prototype behaves as a live, active representation rather than a stored summary waiting to be consulted.

Question 5

A robin is judged a better bird than a penguin. Explain this on the prototype account and on the exemplar account. What can the exemplar account do that the prototype account finds awkward, and why?

On the prototype account, the robin matches the weighted-feature summary of birds closely, while the penguin matches it poorly, so the robin is more typical and verified faster. On the exemplar account, the robin resembles many of the specific birds you have stored, while the penguin resembles few of them. The exemplar account can handle atypical members gracefully, because a penguin can be stored and recognised as itself rather than only as a poor match to an average, and it copes with highly variable categories whose members cannot be sensibly averaged. The prototype account finds these awkward because a single summary discards information about the exceptions and the diversity of the category.

Question 6

Why is the prototype approach better suited to large coherent categories and the exemplar approach to small varied ones? Use furniture in your answer.

In a large, coherent category, members share many features, so an average is informative and storing one summary is efficient: the prototype captures most of what matters. In a small or varied category the members are too diverse to average sensibly, and remembering specific instances preserves information that an average would lose. Furniture is a varied category: what would the average of a wardrobe and a beanbag look like? A single prototype of furniture would be uninformative, whereas stored exemplars (chairs, tables, sofas, wardrobes) let new items be compared with the members they most resemble. People seem to use both strategies, leaning on abstraction when categories are tight and on instances when they are not.

Question 7

Asked what an object is, people say chair rather than furniture or kitchen chair. Explain the information trade-off that makes the basic level special, and why the trade-off is asymmetric.

The basic level (chair) is the level of maximum information for minimum cost. Moving up to the superordinate level (furniture) loses a great deal of information, because members of furniture share far fewer features than members of chair. Moving down to the subordinate level (kitchen chair) adds very little, because a kitchen chair shares almost all its features with chairs in general. The asymmetry, a large loss upward and a small gain downward, makes the basic level the most efficient level at which to categorise, which is why people name objects there spontaneously, children learn it first, and it is the level at which a category can be used in a uniform way.

Question 8

What happens to the basic level with expertise, and why does that finding tell us the privileged level is a fact about the knower rather than about the world?

With expertise the basic level moves down: a birdwatcher names a song thrush as readily and quickly as a novice says "bird" (Tanaka & Taylor, 1991). If the privileged level were a property of the world, of how features are actually distributed across objects, expertise could not move it, because the objects have not changed. That it moves shows the basic level reflects how finely a particular person has learned to carve up a domain, a fact about the knower. Training experts is therefore partly a matter of training new basic-level categories.

Question 9

Collins and Quillian predicted that "a canary has skin" takes longer to verify than "a canary can sing". Explain the prediction and the principle of cognitive economy behind it.

In Collins and Quillian's network, properties are stored at the highest node to which they apply (cognitive economy): "can sing" is stored at CANARY, "can fly" at BIRD and "has skin" at ANIMAL, rather than repeating every property at every node. To verify a statement you traverse links from CANARY to the node holding the property. "A canary can sing" requires no traversal, while "a canary has skin" requires crossing two links, so the

model predicts longer verification times the further the property is stored, and sentence-verification times showed this distance effect.

Question 10

Give two findings the strict hierarchy could not handle, and explain how spreading activation addresses each.

First, typicality: CANARY and OSTRICH are each one link below BIRD, yet "a canary is a bird" is verified faster than "an ostrich is a bird". Second, reversed distance: "a pig is an animal" is verified faster than "a pig is a mammal", although MAMMAL is closer to PIG in the hierarchy (Rips et al., 1973). (In addition, Conrad showed that the distance effect itself was confounded with association strength.) Spreading activation replaces the hierarchy with a network organised by semantic relatedness, in which more closely associated concepts are joined by shorter or stronger links. Canary is more closely linked to bird than ostrich is, so activation reaches BIRD faster. And pig is judged closer in meaning to animal than to mammal, so the link is shorter. Both results become differences in link strength and distance rather than levels in a tree.

Question 11

What is graceful degradation, and why does it make connectionist models attractive as accounts of the brain? Why can a node-based model not produce it as easily?

Graceful degradation means that damage to part of a system degrades its performance gradually and broadly rather than deleting particular items outright. In a connectionist model knowledge is distributed across the weights of many connections, so no fact lives in one place, and removing some units or connections makes everything slightly worse rather than making specific facts vanish. That resembles how real brain damage usually behaves. In a node-based model each concept or fact sits at a particular node, so damaging that node removes that item specifically while leaving the rest intact, which does not naturally produce a gradual, general decline.

Question 12

Connectionist networks usually need many exposures to learn, whereas people sometimes learn a concept in one trial. Explain why this weakness has the same source as the model's strengths.

Connectionist networks learn by gradually adjusting the weights of many connections in response to repeated experience. Because knowledge is spread across shared weights that are changed little by little, the network generalises (learning about one concept informs related ones that share weights) and degrades gracefully (no single weight is critical). The same gradual, distributed weight change means that a single exposure moves the weights only slightly, so many exposures are needed to learn. The strengths and the weakness come from the same property.

Question 13

Set out the similarities and the three main differences between semantic-network and connectionist models, and say which handles typicality better and why.

Similarities: both represent knowledge in interconnected units and both stress the importance of relations between concepts. Differences: (1) structure: semantic networks are organised and often hierarchical, while connectionist networks are distributed and parallel, arranged in layers; (2) where knowledge lives: in labelled nodes and links versus in the weights of connections across many units; (3) learning: semantic networks were specified by the modeller, while connectionist networks acquire their weights gradually from experience. Connectionist models handle typicality better, because typicality emerges naturally as graded patterns of activation, whereas the strict semantic hierarchy could not explain it at all; spreading activation addressed it only by adding graded link strengths.

Question 14

Patients K.C. and E.W. could name cars, tables, tomatoes and pears but not animals. Explain what this pattern shows and why a double dissociation is needed rather than a single case.

The pattern shows a category-specific deficit: knowledge of animals was impaired while knowledge of non-living things and of fruit and vegetables was spared, so conceptual knowledge is not held in one undifferentiated store and different kinds of knowledge depend on partly different tissue. A single case is not enough, because animals might simply be harder to name, for example more visually similar to one another, so any damage would hit them first. The reverse pattern in other patients (Warrington & McCarthy), impaired naming of man-made objects with living things spared, rules out an explanation in terms of difficulty and completes a double dissociation. (This K.C. is not the K.C. of Chapter 4.)

Question 15

Distinguish the sensory-functional, multiple-factor and semantic-category accounts of category-specific deficits. What does it suggest that each explains part of the data and none all of it?

The sensory-functional hypothesis holds that living things are known mainly by their sensory properties and artifacts mainly by their function, so damage to sensory stores impairs living things and damage to functional stores impairs artifacts. The multiple-factor approach says concepts are distinguished along several dimensions at once, with animals weighted towards motion, colour and size and artifacts towards actions performed with them. The semantic-category approach proposes dedicated neural circuits for evolutionarily important categories such as animals and plant foods. The sensory-functional account struggles with K.C. and E.W., whose fruit and vegetable naming, things known largely by appearance and taste, was spared. That each account explains part of the data and none all of it suggests the organising principle may be more general than any of them, consistent with a model in which knowledge depends on several modality-specific regions bound together.

Question 16

Give one piece of evidence for the embodied approach, and state the strongest objection to it. How does weak embodiment respond?

Evidence: when people read action words such as lick, pick and kick, they differentially engage the face, hand and leg regions of motor cortex in the same somatotopic order as the actions themselves (semantic somatotopy; Hauk et al., 2004). Mirror neurons, which fire both when an action is performed and when it is observed, offer a possible mechanism. The strongest objection concerns abstract concepts, such as justice or prime number, which are hard to ground in sensorimotor experience; in addition, mirror neurons may accompany understanding rather than contribute to it. Weak embodiment accepts that concepts are grounded in perception and action but allows that some knowledge is genuinely abstracted away from the body, which is the position the hub-and-spoke model takes.

Question 17

rTMS over the anterior temporal lobe slows naming for all categories; over the parietal lobe it slows only tools. Explain how this pattern identifies the hub and the spoke, why the identification does not have to be assumed in advance, and what control shows that the anterior temporal effect is semantic rather than a general slowing of speech.

Pobric and colleagues found that rTMS over the left anterior temporal lobe slowed naming of both living and man-made things, a category-general effect, while rTMS over the left inferior parietal lobe slowed naming only of manipulable man-made objects such as tools. If all concepts route through a hub, disrupting it should impair every category; disrupting a spoke should impair only the concepts that depend on it, here knowledge of how objects are manipulated. The hub is identified by which disruption is category-general and specifically semantic, so it is not assumed in advance. Category-general slowing could also arise from disrupting a general stage of naming, such as retrieving or articulating words. The control task showed anterior temporal stimulation slowed

naming pictures, which requires retrieving a concept, but not naming numbers, which requires the same spoken output with little conceptual retrieval, so the effect is semantic.

Question 18

Why is the hub-and-spoke model described as a middle position rather than a rival to the embodied approach?

The hub-and-spoke model accepts the central embodied insight: the spokes are modality-specific regions for vision, motion, sound, action, words, emotion and taste, so concepts remain grounded in perception and action. It adds a transmodal hub in the anterior temporal lobes that binds the spokes into coherent concepts that can transfer across modalities and generalise across members, which is what abstraction requires and strong embodiment struggles to deliver. It therefore absorbs the grounding claim rather than rejecting it, committing to weak embodiment, and its evidence counts against strong embodiment as much as against fully disembodied symbols.

Question 19

Using typicality and the basic level, explain both why an experienced clinician is fast and accurate and why they are most at risk of missing an atypical case. Why is this not a failure of diligence?

Expertise moves the basic level downward, so the experienced clinician recognises a specific condition as quickly and fluently as a novice recognises a coarse category, matching the case to a rich stored prototype or set of exemplars. Because that matching runs on typicality, typical presentations are recognised quickly and correctly. The same mechanism means atypical presentations, such as a heart attack without chest pain, match the prototype poorly and are more likely to be missed or misfiled, and the expert's fast, fluent judgement makes this harder to notice. This is not a failure of diligence: it is the cost of the mechanism that makes expert judgement fast and usually right. A good clinician learns to distrust the fluent prototypical match when the stakes are high.

Question 20

Explain how a prototype becomes a stereotype, and say what is and is not different about the cognitive processes involved.

When the category is a kind of person, the prototype is a weighted summary of features thought typical of that group, and applying it to individuals is a stereotype: it speeds judgement and favours the typical member while resisting atypical or borderline ones, just as with birds. The cognitive processes are the ordinary ones of categorisation (graded similarity, typicality and fast automatic matching), so nothing special is needed to produce a stereotype. What is different is the object of categorisation: the people being categorised have an interest in the outcome that chairs and birds do not, so the same mechanism has social and ethical consequences.

Chapter 8: Visual Imagery

Question 1

Distinguish visual from spatial imagery, and explain why a congenitally blind person is the clearest evidence that the two are separable.

Visual imagery represents how something looks: its shape, colour and visual detail. Spatial imagery represents where things are and how they are arranged, their locations and relations, without necessarily any visual quality. A congenitally blind person has never had visual experience and so has no visual imagery, yet can have excellent spatial imagery, reasoning fluently about layouts and routes. One capacity is entirely absent while the other is intact, which shows they are separable. They also lean on partly different streams: the ventral "what" pathway for appearance and the dorsal "where" pathway for spatial relations.

Question 2

Why does the distinction between visual and spatial imagery complicate the claim that mental rotation shows images are "picture-like"?

Mental rotation shows that transforming an image takes time in proportion to the angle, as if a real object were being rotated. But rotation is a transformation of spatial structure: it depends on the arrangement and orientation of parts, not on colour, texture or other visual detail. The result is therefore strictly evidence about spatial imagery, and could be produced by a spatial representation with no visual, picture-like quality at all. Milton and colleagues found that rotation and scanning tasks discriminate poorly between aphantasic and hyperphantasic people, which confirms that these tasks measure spatial imagery rather than visual experience.

Question 3

Explain Paivio's conceptual-peg account, and say why concrete nouns should benefit from it more than abstract ones.

Paivio found that concrete nouns that readily evoke an image, such as magician, were remembered far better in paired-associate learning than abstract nouns such as discourse. On his conceptual-peg hypothesis, a concrete noun evokes an image that acts as a peg on which its associate can be hung. His later dual-coding theory proposed that a concrete word is stored twice, verbally and as an image, so retrieval has two routes instead of one. Concrete nouns benefit more because they readily evoke images, whereas abstract nouns have no obvious image, so they are stored mainly in the verbal code and lack the extra route.

Question 4

Shepard and Metzler found reaction time rose linearly with the angle of rotation. Explain why the depictive account predicts this, and why the linearity matters more than the slowing.

On the depictive account, a mental image represents space with space, so rotating it means transforming it continuously through intermediate orientations, as with a real object. If this happens at a constant rate, the time required should rise linearly with the angle to be traversed, which is what Shepard and Metzler found. The linearity matters more than the slowing because a propositional system could be slower for larger angles for many reasons, for example because the problem is harder. What it does not obviously predict is a constant rate of transformation, which is the signature of an analogue process passing through intermediate states.

Question 5

Why does a propositional list of facts fail to predict the image-scanning result of Kosslyn, Ball and Reiser? Be precise about what a list lacks.

Participants memorised a map of a fictional island, imagined it and mentally scanned from one landmark to another; scanning time rose in direct proportion to the real distance between landmarks. A propositional list

such as "the hut is north of the well" encodes relations between landmarks but not metric distance, and a list is not traversed at a rate, so it contains nothing that would make one scan take longer than another in proportion to distance. A representation that preserves metric spatial structure, so that moving between distant points means crossing more intervening space, predicts exactly the observed timing.

Question 6

State Pylyshyn's tacit-knowledge critique and explain why it cannot be defeated by behavioural evidence alone. What kind of evidence does it require instead?

Pylyshyn argued that imagery results need not reflect the format of the representation. Participants know how real objects behave and may unconsciously reproduce that knowledge, taking longer to rotate through larger angles or to scan longer distances because they know these take longer in reality, not because anything is being rotated or scanned. This cannot be defeated by behaviour alone because any timing pattern a depictive representation would produce is also one a well-informed participant could produce by simulating what they expect. It requires evidence that tacit knowledge could not fake, such as neural evidence: a retinotopic map recruited according to imagined size, a causal effect of disrupting visual cortex, or a reflex such as the pupillary light response to imagined brightness.

Question 7

Give one respect in which imagery resembles perception and one in which it clearly differs, and relate the difference to the Rider and the Elephant.

Resemblance: imagery behaves like perception in fine detail. Details of an imagined object are easier to see when it is imagined large, just as with real viewing, and imagining a letter can prime its later recognition.

Difference: perception is automatic and stable, while imagery is effortful and fragile, must be actively generated and maintained, and fades as soon as attention lapses; images are also harder to reinterpret than percepts. In the book's terms, perception is the Elephant's effortless delivery, supplied at no cost whether you attend or not, while sustained imagery is the Rider's laborious construction, which is why it is tiring in a way that looking is not.

Question 8

Explain how the TMS result of Kosslyn and colleagues (1999) does something the fMRI result of Le Bihan and colleagues (1993) cannot.

Le Bihan and colleagues showed that primary visual cortex is active during both perception and imagery, but that is a correlation: it shows the region is engaged, not that imagery depends on it, and a participant simulating what they expect could also produce activity in visual areas. Kosslyn and colleagues applied disruptive TMS to early visual cortex and slowed both perception and imagery. Interfering with the region impaired imagery, so the region is doing part of the work: the contribution is causal. If imagery were purely propositional, disrupting visual cortex should not have slowed it.

Question 9

What does it mean to say early visual cortex is recruited retinotopically during imagery, and why is that pattern difficult for a propositional account to explain?

Early visual areas are retinotopic: adjacent points in the visual field map to adjacent points on the cortex, with the centre of gaze at the occipital pole and the periphery further forward. Kosslyn and colleagues (1995) found that imagining a small object activated the posterior, foveal part of V1, while imagining a large object filling the mind's eye activated more anterior regions representing the periphery. So the cortical map is used in a size-appropriate way, as in perception. A propositional account struggles with this because there is no obvious reason why a list of descriptions should be laid out across a retinotopic map according to an imagined object's size.

Question 10

Patient M.G.S. could mentally approach a horse to within fifteen feet before surgery and only thirty-five feet afterwards. Explain why a smaller visual cortex produces a larger overflow distance rather than a smaller one.

The mind's eye has a limited field, and an imagined object overflows it when it fills more than that field. Removing the right occipital lobe reduced the amount of visual cortex and so shrank M.G.S.'s imagined visual field. A smaller field is filled by a given object sooner, which happens at a greater imagined viewing distance. The distance at which the image overflows therefore became larger, from about fifteen to about thirty-five feet. The result ties the size of the mind's eye to the amount of visual cortex, in one patient measured twice.

Question 11

R.M. could draw an object in front of him but not from memory; C.K. could not recognise objects but could draw them from imagery. What does this double dissociation show, and what would a single case have failed to establish?

R.M. had perception without imagery: he could draw an object placed in front of him but not from memory. C.K. had imagery without perception: with visual agnosia he could not recognise objects, even drawings he had made himself, yet could draw detailed objects from imagery. Each capacity is lost in one patient and spared in the other, so imagery and perception cannot depend on exactly the same mechanisms, though they share a good deal. A single case, such as R.M. alone, could be explained by imagery simply being the more demanding or fragile task, so that any damage to a shared system would impair it first. The reverse pattern in C.K. rules out that explanation.

Question 12

In Bisiach and Luzzatti's patients, the side of the square that was omitted changed with the imagined vantage point. Why does that show the knowledge was intact and the deficit was one of attention?

Asked to imagine a familiar square from one end, the patients described only its right-hand side; imagining it from the opposite end, they described the side they had just omitted and omitted the side they had just described. Both halves of the square were reported at some point, so the knowledge of the whole square was plainly intact. What determined what was omitted was the left side relative to the imagined viewpoint, so the deficit was in attending to one side of a representation, not in the stored knowledge. The representation being neglected was an image.

Question 13

What is aphantasia, and why is it more informative for theory than a simple report of low imagery vividness would be?

Aphantasia is the congenital absence of voluntary visual imagery (Zeman et al., 2015), reported by about four per cent of people: an aphantasic person can describe a banana's colour and shape perfectly well but reports seeing nothing in the mind's eye. It is more informative than a report of low vividness because it separates having visual information from having visual experience: the information is evidently present while the experience is absent. That makes it possible to ask what is missing, and to find, for example, that visual-area activity is largely preserved while connectivity is reduced. It also has an objective marker, the absence of the imagery pupillary light response, rather than resting on self-report alone.

Question 14

Milton and colleagues found that aphantasic, mid-range and hyperphantasic groups performed equally on standard laboratory memory tests but differed markedly on autobiographical memory and imagination. Explain what that combination rules out, and connect it to what §4.4 said about episodic memory and the self.

Equal performance on standard laboratory recall and recognition tests rules out the idea that aphantasia is a general memory impairment. The graded differences (hyperphantasia above controls above aphantasia) on

autobiographical memory and imagination, including imagining one's own future, show that what varies is the kind of remembering that requires projecting yourself into a constructed scene. Section 4.4 identified that auto-noetic, self-locating recollection as the distinctive feature of episodic memory, and Chapter 4 suggested that remembering the past and imagining the future share machinery. The imagery literature arrives at the same conclusion: recollection and imagination share a system, and the sensory vividness of that system varies between people.

Question 15

Why is a three-group design comparing both extremes against a mid-range control stronger evidence than comparing aphantasic participants with everyone else?

Comparing aphantasic participants with everyone else treats aphantasia as a deficit and shows only that one group differs from the rest, which could reflect something peculiar to that group. Comparing both extremes with a mid-range control tests the continuum itself: if hyperphantasic participants outperform controls, who outperform aphantasic participants, the relationship is graded and runs through the whole range of vividness. A graded, ordered pattern across three groups is much harder to explain by an incidental feature of one group, and it shows vividness is a spectrum rather than a dichotomy of normal versus impaired.

Question 16

Classic mental-rotation and scanning tasks discriminate poorly between aphantasic and hyperphantasic people. Why is that awkward for using those tasks as measures of visual imagery, and which distinction from §8.1 does it vindicate?

If mental rotation and scanning measured visual imagery, people who report no visual experience at all should perform them very differently from people whose imagery is nearly perceptual. They perform much alike, so these tasks cannot be good measures of visual imagery experience. This vindicates the visual-versus-spatial distinction of §8.1: rotation and scanning draw on spatial imagery, which aphantasia can spare, rather than on visual imagery. It is why objective measures of visual imagery, such as the pupil response, were needed.

Question 17

Milton and colleagues measured connectivity at rest; Liu and colleagues measured it during an imagery task. Explain why agreement between two such different designs strengthens the conclusion more than a larger sample in either study alone would.

The two studies differ in samples, scanners, field strength and, above all, condition: Milton and colleagues measured connectivity at rest with no task, finding stronger prefrontal-to-visual connectivity in hyperphantasia, while Liu and colleagues measured it during imagery, finding reduced connectivity between the Fusiform Imagery Node and frontoparietal areas in aphantasia. Each design has its own weaknesses and possible confounds. When designs with so little in common point to the same kind of frontal-to-visual coupling, it is very unlikely that a shared artefact explains both. A larger sample in one study would reduce random error but would not remove that study's systematic biases. Convergence matters especially because Liu and colleagues had only ten participants per group.

Question 18

In aphantasia, visual areas are active during imagery but connectivity between the Fusiform Imagery Node and frontoparietal networks is reduced. Why does this support the view that experience depends on integration rather than local activation? What does it imply about V1, and about the retinotopic findings in §8.4?

If local activation of visual areas were sufficient for the experience, aphantasic people, whose visual-area activity during imagery is largely present, should have the experience. They do not, and what distinguishes them is reduced communication between the fusiform hub and frontoparietal networks. Experience therefore seems to depend on high-level visual cortex being integrated with frontoparietal networks, not on any one area being active: the parts are intact but the conversation between them is not (a functional disconnection). For V1,

aphantasic participants show stronger deactivation of foveal V1, and patients with V1 lesions can retain vivid imagery, so V1 is not necessary for the experience. The retinotopic findings of §8.4 should therefore be read as showing that V1 contributes to imagery, especially vivid and detailed imagery, not that it generates the experience.

Question 19

How can the pupil be used as an objective index of imagery, what would you expect in an aphantasic participant, and why is a reflex better evidence here than a vividness rating?

The pupil constricts reflexively to brightness, and Kay and colleagues found it also constricts more when people imagine a bright object than a dark one, so the difference between imagined-bright and imagined-dark trials indexes imagery. In an aphantasic participant you would expect no difference: the bright and dark imagery conditions should not separate, while the pupil still responds normally to real light and dilates with cognitive load. A reflex is better evidence than a vividness rating because the participant cannot produce it to order or suppress it to be helpful. It sidesteps the problem that vividness ratings are self-reports about something nobody else can check.

Question 20

In Kay and colleagues' task an eight-second black screen separated perception from imagery. What objection does that interval rule out, and why would the result be much weaker without it?

Without the gap, the pupil might still be responding to the real stimulus when the imagery period began, and any brightness-dependent difference during imagery could be dismissed as a lingering after-image or a residual perceptual response. Eight seconds of black screen let any after-image fade and the pupil settle before imagery started. Without that interval, the critical result could have been explained without any imagery at all.

Question 21

The imagery pupil response is several times smaller than the perceptual one. Does that weaken the claim that imagery recruits perceptual machinery? Argue both ways.

It weakens the claim if "recruits perceptual machinery" is taken to mean imagery is simply perception at reduced strength: during perception the bright and dark conditions separate by about three quarters of a millimetre, while during imagery they separate by about two tenths, so imagery does only a small fraction of what light does, consistent with other evidence that imagery and perception share some but not all machinery. It does not weaken it if the claim is that imagery engages the same pathway: a reflex that responds to imagined brightness at all, in a direction that matches perception and scales with trial-by-trial vividness, shows that the imagery signal reaches perceptual machinery. A smaller effect is what one would expect from an internally generated signal that is weaker than real input.

Question 22

The pupil response tracked trial-by-trial vividness ratings and an objective rivalry measure, but not questionnaire scores or average vividness ratings. What does that pattern suggest about the limits of questionnaire measures of inner experience?

The pupil response agreed with measures taken trial by trial within a person (vividness ratings for individual images) and with an independent objective measure (binocular rivalry), but not with between-person self-report measures (questionnaire scores or average vividness across the experiment). This suggests questionnaires may not capture inner experience accurately, perhaps because people have no common scale for comparing their imagery with others', or because general self-descriptions differ from moment-to-moment reports. Studies that rest on people rating their own inner states should be treated with caution, and objective measures are needed.

Question 23

Aphantasic participants showed no imagery pupil response but normal perceptual pupil responses and normal load-related dilation. Explain why those two control findings are what turn an absence into evidence, and what could have been concluded without them.

Without controls, an absent imagery pupil response could be explained trivially: aphantasic participants might have an impaired pupillary reflex, or they might not have been engaging with the task. Normal pupillary light responses to the real stimuli show that their pupils and reflex arc are intact. Normal dilation with greater cognitive load (four versus one triangle) shows they were engaging with the task. With those ruled out, the absence is specific to imagery, which turns a null result into positive evidence that a reflex that tracks imagined brightness in everyone else does not do so in aphantasia, the first physiological validation of the condition. Without the controls, nothing specific about imagery could have been concluded.

Question 24

Explain what motor imagery adds to the case that imagery reuses the machinery of perception and action. What does the relationship between imagined effort and corticospinal excitability show?

Motor imagery, imagining a movement without making it, obeys the timing laws of real movement, so imagined and executed actions take similar durations, and it engages much of the same preparatory motor machinery. It extends the case beyond vision to action, suggesting that imagery generally is simulation using the systems built for perceiving and acting. TMS shows that motor imagery raises corticospinal excitability in primary motor cortex in proportion to imagined effort: imagining a forceful contraction drives the motor pathway more strongly than imagining a weak one (Helm et al., 2015). The imagined force is written into the excitability of the motor pathway, so imagining an action partly prepares it, which is why mental rehearsal is useful in sport and rehabilitation.

Question 25

Pearson and Kosslyn say the imagery debate should end. What do they mean, whose position is vindicated, and how does aphantasia fit their conclusion?

They mean the question posed as a winner-take-all contest, pictures or propositions, is the wrong one. People use both formats and switch between them flexibly depending on the task. Both positions are partly vindicated: Kosslyn's, because depictive representations are real, functional and grounded in the visual system; and Pylyshyn's, because they are not the only currency of thought. The productive question is which format is recruited when, by whom and for what task. Aphantasia fits because an aphantasic person reasons about visual facts perfectly well by a non-depictive route. If depictive representation were the only way to hold visual knowledge, aphantasia would be a knowledge deficit; it is not.

Question 26

Explain why the method of loci works, using at least two ideas from this chapter and one from Chapter 5.

The method of loci places vivid images of the items at successive locations along a familiar route and retrieves them by mentally walking the route. It forces elaborative, imageable encoding, giving each item a distinctive image (Paivio's imagery advantage). It recruits spatial as well as visual imagery, so the route provides an organised retrieval structure that preserves order. From Chapter 5: walking the route supplies a self-generated retrieval cue for each item, and self-generated cues are far more effective than cues supplied by others (Mäntylä); it also reflects organisation at encoding as an investment in retrieval, and encoding specificity, since the same route is reinstated at recall.

Question 27

Why must a pegword image be interactive rather than simply two images side by side?

The peg works by providing a cue that retrieves the item. If the bun and the hat are imagined side by side, the two are stored as separate images with no link between them, so retrieving the peg (bun) does not bring back

the item (hat). An interactive image, a bun wearing a hat, binds the two into a single integrated representation, so activating the peg retrieves the item. Integration, not mere co-occurrence, creates the association.

Question 28

The place-based technique studied by Reser and colleagues adds narrative to location. What does narrative contribute that place alone does not?

Place supplies an organised set of locations, a spatial retrieval structure, but the items at successive locations are otherwise unconnected. Narrative adds causal and temporal structure: a story links each item to the next through events, so recalling one part of the sequence leads to the next through the story's logic. That is a second organising principle on top of place, giving memory more and more distinctive holds on the material. Reser and colleagues found the Aboriginal place-based narrative technique at least as effective as, and on some measures better than, the classical method of loci.

Chapter 9: Language

Question 1

Explain what is meant by saying language is hierarchical, rule-governed and arbitrary, giving an example of each.

Hierarchical: a small set of elements (sounds, then words) combines without limit into larger units (phrases, then sentences), so a finite vocabulary yields an unbounded number of sentences; for example, "the cat" can be embedded in "the cat that the dog chased", and so on without end. Rule-governed: combinations follow a grammar, so "the cat chased the dog" means something different from "the dog chased the cat", and "chased dog the cat the" means nothing. Arbitrary: apart from a few onomatopoeic words, a word bears no resemblance to its referent; there is nothing dog-like about the sound "dog", and a community of speakers simply agrees on it.

Question 2

What observations about children's language did Chomsky use against Skinner, and what did he conclude? What did the debate not settle?

Chomsky observed that children produce sentences they have never heard, and that were therefore never reinforced, and that they acquire complex grammatical rules from input too impoverished to specify those rules (the poverty-of-the-stimulus argument). (Children also make grammatical errors that were never reinforced.) He concluded that the core of language is substantially innate. The critique was decisive against a pure reinforcement account and reoriented the field towards the mental machinery of language. It did not settle how much of that machinery is language-specific rather than general-purpose learning, a question that statistical-learning findings have reopened.

Question 3

How does statistical learning in infants bear on the poverty-of-the-stimulus argument?

Saffran and colleagues showed that eight-month-olds who heard two minutes of a continuous stream of nonsense syllables could later distinguish its "words" (syllables that reliably co-occurred) from sequences spanning word boundaries. Infants track transitional probabilities and use dips in predictability as cues to boundaries. This bears on the poverty-of-the-stimulus argument because it shows the input is richer than it looks once its statistics are counted, and that infants extract more structure from it, quickly, than was assumed. Whether statistical learning can reach the grammatical knowledge at the heart of the argument is still debated, so it weakens rather than refutes it.

Question 4

What is the word-frequency effect, and how do eye-movement data demonstrate it?

The word-frequency effect is that people respond faster to common words than to rare ones; for example, they name pictured objects faster when the names are common (Wingfield, 1968). Eye-tracking during reading shows it directly: readers fixate low-frequency words for longer than high-frequency words (Rayner & Duffy, 1986). Fixation duration gives a moment-by-moment measure of processing during ordinary reading, without asking the reader to make any judgement.

Question 5

Why is variable pronunciation a problem for any account of word recognition based on matching the signal to a stored template?

In running speech, words do not appear as clean instances of stored forms: "did you" surfaces as "did-joo" and "going to" as "gonna", and pronunciation varies across speakers, accents and contexts. A template-matching

account needs the incoming signal to match a stored form; because the signal often does not, recognition must rely on knowledge brought to the signal, such as context and expectations about what is likely to be said, rather than on matching alone.

Question 6

On first meeting an ambiguous word, do we access one meaning or several? Cite the evidence, and explain the role of meaning dominance.

We briefly access several. Tanenhaus and colleagues (1979) found that on first encountering an ambiguous word, people activate all its meanings, shown by the word priming responses to words related to any of its senses; only afterwards does context prune away the irrelevant meaning. Which meaning wins, and how fast, depends on meaning dominance. With biased dominance, one meaning is far more frequent and is reached quickly unless context forces the rarer one. With balanced dominance, the meanings are about equally frequent, so without context the reader is briefly slowed while they compete. Meaning resolution is a race between frequency and context.

Question 7

Explain why "While the man hunted the deer ran into the woods" is a garden-path sentence, and what late closure contributes to the explanation.

Parsing is incremental, so we group words into phrases as they arrive. On reaching "the deer" we attach it as the object of "hunted" (the man hunted the deer). At "ran" that parse fails, because "ran" needs a subject, and we must backtrack and reinterpret "the deer" as the subject of a new clause ("the deer ran into the woods"). The sentence is only temporarily ambiguous, and the measurable slowdown at "ran" is the cost of reanalysis. Late closure is the heuristic of attaching each new word to the phrase currently being built, which is why "the deer" is bundled into the hunting clause in the first place, leading the reader up the garden path.

Question 8

How did Tanenhaus and colleagues (1995) use the visual world to distinguish the garden-path model from the constraint-based approach? Why is the timing of the context effect, rather than its existence, what settles the question?

Listeners followed instructions such as "Put the apple on the towel in the box" while looking at a real scene. With one apple in the scene, they briefly looked at an empty towel, having parsed "on the towel" as the destination. With two apples, one on a towel, they did not, because they used the scene to parse "on the towel" as specifying which apple. Both models agree that context eventually influences interpretation. The garden-path model says syntax alone guides the initial parse and context only repairs it afterwards; the constraint-based approach says all information, including non-linguistic context, guides the parse immediately. The eye movements showed context shaping the parse within a fraction of a second, which is immediate guidance, so the timing rather than the mere existence of a context effect distinguishes the two.

Question 9

What is a situation model, and why is it, rather than the exact sentences, what we tend to remember?

A situation model is a mental representation of the state of affairs a text describes, rather than of its words, built from the protagonist's point of view and filled in by inference and schemas. Readers represent the story almost as if experiencing the situation themselves. It is what survives because comprehension aims at what the text is about: the exact wording is discarded once the meaning is extracted and integrated, while the situation model is connected to existing knowledge and so is more durable.

Question 10

Explain why two readers with different cultural backgrounds can build different situation models from the same text without either misreading it.

Building a situation model depends on inference, and inference draws on the schemas a reader actually has. A text that assumes knowledge one reader shares and another does not will support inferences for the first but not the second, and each reader imports their own schemas, filling gaps with what would be normal in their own world. Both can read the words accurately, and even recall them accurately, while building different models of the situation. This is a difference in background knowledge, not a failure of language or intelligence, which is why a comprehension test is partly a test of shared background.

Question 11

Explain syntactic priming, and why the fact that the second picture is different from the first matters to the interpretation.

Syntactic priming (syntactic coordination) is the tendency to reuse a grammatical construction just produced or heard. Branigan and colleagues had a confederate describe a card with one of two grammatical forms, after which participants described a new card and tended to reuse the structure they had heard. It matters that the second picture was different because, if it had been the same, participants could have repeated the confederate's words or content. Reuse of the structure with different content shows that the grammatical form itself was primed, independently of meaning and words, which reduces the computational load of planning the next sentence.

Question 12

State the double dissociation between Broca's and Wernicke's aphasia, and give two reasons the classical two-area picture is a simplification.

Broca's aphasia, following damage to Broca's area in the left frontal lobe, leaves speech laboured and agrammatical while comprehension is relatively intact (except for syntactically complex sentences). Wernicke's aphasia, following damage to Wernicke's area in the left temporal lobe, leaves speech fluent but empty while comprehension is poor. Production and comprehension are therefore separable. It is a simplification because, first, language depends on a distributed network including the connections between these regions, and real aphasias rarely fall neatly into two types; and second, the mapping to the left hemisphere, while typical, is not universal.

Question 13

Why is it a consequential error, rather than merely a discourtesy, to treat Aboriginal English as incorrect English? Set out the clinical consequence.

Aboriginal English is a distinct, rule-governed dialect with its own systematic grammar, so a speaker is following rules rather than failing to follow Standard Australian English ones. Treating it as incorrect is a measurement error with real consequences. A standardised language assessment written for Standard Australian English counts dialect features as errors and can score a fluent Aboriginal English speaker as impaired. Children can then be referred for clinical intervention they do not need, while genuine language difficulties go unnoticed because the assessment is not measuring what it claims to measure.

Question 14

Name four dimensions along which bilingual speakers differ, and explain why a study recruiting by a single yes-or-no question may reach a different conclusion from another doing the same.

Bilingual speakers differ in the age and context at which they acquired each language; in proficiency in each; in how often and in which settings they use each; in how similar their languages are; in the relative social prestige of the languages; and in their emotional relationship to each. (Any four will do.) A single yes-or-no question collapses all of this into a binary, so two studies using the same label may recruit very different people: one may

get highly proficient speakers who switch frequently, the other people with rarely used schoolroom French. If an effect depends on particular aspects of bilingual experience, the studies will disagree for that reason alone.

Question 15

Distinguish simultaneous from sequential bilingual acquisition, and say what different problem each route faces.

Simultaneous bilinguals acquire two languages from the start; sequential bilinguals add a second language after a first is established. A simultaneous learner must sort two systems apart while building both at once. A sequential learner already has one system in place, which can help (transferring knowledge and skills) but can also interfere with the new language.

Question 16

Why was codeswitching once read as confusion, and what is the evidence that it is instead a marker of competence? What does it have in common with the Aboriginal English case?

Moving between languages was taken as a sign that a speaker lacked full mastery of either language and was mixing them because they could not keep them apart. The evidence now shows codeswitching is rule-governed: it tends to occur at particular syntactic boundaries, and doing it well requires fluency in both languages (Yow et al., 2018). Like Aboriginal English, it is a case where a deficit reading was imposed on what is really a competence: systematic, rule-following language use was mistaken for error because it was judged against a monolingual standard.

Question 17

The bilingual executive-function advantage replicates inconsistently. Give the most promising explanation for the inconsistency, and explain why a larger sample would not resolve it.

The most promising explanation is that bilingualism is a graded, multidimensional profile rather than a trait, while studies typically recruit by a yes-or-no question. If only some profiles (high proficiency, frequent switching, immersion in a community using both languages) are associated with better executive control, studies that happen to recruit those people will find an effect and those that recruit others will not, even if every study is competently run and honestly reported. A larger sample would only estimate the average of that heterogeneous mixture more precisely; it would not separate the profiles that matter from those that do not. The remedy is to measure the dimensions of bilingual experience and model them.

Question 18

A meta-analysis finds that second-language learners achieve better mathematics results. Give one mechanistic explanation and one confound, and say what evidence would distinguish them.

Mechanism: mathematics draws on executive abilities such as working memory, inhibition and flexibility, which managing and switching between two languages is thought to train. Confound: students who take up a second language are not a random sample. They may differ in prior ability, motivation, family socioeconomic background or school resources, any of which could produce both language study and better mathematics results. To distinguish them you would need designs that control selection: random assignment to language learning, or longitudinal studies measuring mathematics before and after starting a language and controlling for prior attainment and background, and ideally evidence that gains are mediated by improvements in executive function.

Question 19

Explain, using processing fluency, why a foreign accent can reduce the credibility of a claim, and why brief exposure to the accent reduces that effect. What does this imply about how to counter the bias?

Foreign-accented speech is harder to process, harder processing feels like doubt, and that doubt is attributed to the claim rather than to the accent, so the claim seems less credible, the same fluency mechanism that drives the illusory-truth effect. Brief exposure to the accent makes it easier to process, which reduces the disfluency

and so reduces the credibility penalty (Boduch-Grabka & Lev-Ari, 2021), as the fluency account predicts. (The original effect has a mixed replication record.) Because the bias arises automatically in processing, below deliberate judgement, it is not reliably corrected by trying to be fair. It is better countered by changing the conditions: increasing exposure to different accents, using structured assessment, and not relying on impressions of credibility where a decision matters.

Chapter 10: Problem Solving and Creativity

Question 1

Distinguish well-defined from ill-defined problems, and explain why the information-processing approach suits the former better. Why does the nesting of well-defined sub-problems inside ill-defined ones matter for the debate between the two accounts?

Well-defined problems have a clearly specified starting state, a finite set of legal moves (operators) and an unambiguous goal, such as an algebra equation or the Tower of Hanoi. Ill-defined problems lack one or more of these, such as "paint a beautiful picture", which has no agreed goal and no fixed operators. The information-processing approach models solving as searching a problem space from start to goal using operators, so it needs those elements to be specified, which is true of well-defined problems. The nesting matters because most real and creative problems are ill-defined but contain well-defined sub-problems. The search account describes the nested sub-problems well, and the Gestalt restructuring account describes the work of deciding what the sub-problems are, so the two accounts divide the territory rather than compete for it.

Question 2

Explain the difference between an algorithm and a heuristic, and what each buys and costs.

An algorithm is a step-by-step procedure guaranteed to produce the correct result for a well-defined problem, like a recipe. It buys certainty, but it must be followed exactly and for large problem spaces it can be impossibly slow. A heuristic is a rule of thumb that often works but is not guaranteed, such as working backwards or breaking a task into subgoals. It buys speed and economy, and is used when there is too much information, too little time, too little information or little at stake, but it costs reliability: its occasional failures are the source of systematic biases.

Question 3

What is means-ends analysis, and how do subgoals help solve the Tower of Hanoi? Why does exhaustive search fail for the travelling salesperson problem?

Means-ends analysis is a heuristic in which, at each step, you identify the largest difference between your current state and the goal and choose an operator that reduces it; if no single move can reduce it, you set a subgoal, a nearer state that makes the reducing move possible. In the Tower of Hanoi you cannot move the largest disc until the smaller ones are elsewhere, so you set the subgoal of moving the smaller discs to the middle peg, move the largest disc to the third peg, then rebuild on top of it; applied recursively this solves the three-disc puzzle in seven moves. Exhaustive search fails for the travelling salesperson problem because the number of possible routes explodes so fast with the number of cities that checking them all is infeasible, even though the rules are clear. People nonetheless find near-optimal routes, apparently by perceiving global spatial structure.

Question 4

Describe Metcalfe and Wiebe's (1987) warmth-rating result. Why is asking people repeatedly how close they are better evidence than asking them whether insight felt sudden?

Participants worked on insight problems and non-insight algebra problems and rated every fifteen seconds how warm, or close to a solution, they felt. For algebra problems warmth rose steadily as they approached the answer; for insight problems it stayed low and flat until it jumped in the final interval before the solution. Asking whether insight felt sudden would yield a report about a feeling, which could reflect expectations or how the experience is remembered. Repeated warmth ratings can be compared with actual progress, and showed that people's metacognition tracked progress for non-insight problems but not for insight problems. That is a behavioural difference between problem types, not merely a reported one.

Question 5

State the deflationary interpretation of insight, and say what the flat warmth curve would mean on that reading.

The deflationary interpretation is that all problems are solved by unconscious incremental processing, and the insight "Aha!" is simply the moment that processing crosses into awareness, not a distinct mechanism. On that reading the flat warmth curve does not mean no progress was being made; it means the solver had no conscious access to the progress being made unconsciously. In the book's terms, the incremental work is the Elephant's and the Rider meets the answer only when it arrives.

Question 6

Köhler read Sultan's behaviour as insight. How do the circumstances of Sultan's two-stick solution, and Birch's (1945) finding on prior experience, complicate that interpretation?

Köhler read Sultan's abrupt, purposeful stacking of boxes or joining of two sticks as insight rather than trial and error. But by Köhler's own account the two-stick solution came after an hour of failed attempts, while Sultan was idly handling the sticks, so it may have been reached through manipulation rather than sudden restructuring. Birch found that chimpanzees with no prior experience of playing with sticks rarely solved stick problems at all. On this evidence insight may build on earlier trial-and-error experience rather than replace it.

Question 7

Explain, using the mutilated-checkerboard problem, why a change of representation can turn an intractable problem into an obvious one. What information does the useful representation discard, and why is discarding it helpful?

Remove two diagonally opposite corners from a 64-square board, leaving 62, and ask whether 31 dominoes, each covering two adjacent squares, can tile them. Represented by position, the problem invites endless attempts at arrangements. Represented by colour, it is obvious: every domino covers one light and one dark square, but the two removed corners are the same colour, so the board has 32 squares of one colour and 30 of the other, and no tiling can succeed. The useful representation discards almost all the detail people fixate on, the whole geometry of the board, and keeps one fact, the colour count. Discarding detail helps because the irrelevant information was what made the search seem necessary and endless; a good representation is more selective rather than richer.

Question 8

What is functional fixedness, and how do the candle and two-string problems demonstrate it?

Functional fixedness is the difficulty of using an object for anything other than its usual function, a special case of mental set. In Duncker's candle problem, people must fix a candle to a wall with a box of tacks and matches; they struggle because they see the box as a container for tacks rather than as a platform that can be tacked to the wall. In Maier's two-string problem, two strings hang too far apart to grasp both; the solution is to tie the pliers to one string and swing it as a pendulum, which is hard because the pliers are seen as a tool rather than a weight.

Question 9

Using Table 10.1, explain how problems 2 to 6 create the mental set, why problems 7 and 8 fail to break it even though a two-pour route exists, and why problem 9 exposes it. What does the untrained control group contribute to the conclusion?

Problems 2 to 6 are all solved by $B - A - 2C$, and for most of them there is no shorter route, so five consecutive successes make that routine simply what a water-jug problem is. In problems 7 and 8 a two-pour route exists ($A - C$, $A + C$), but the long method also gives the right answer, so nothing signals that anything simpler is available, and being right is mistaken for being efficient. In problem 9 the long method fails (it gives 42, not 25), and only $A - C$ works, so the set becomes visible as participants fail, take far longer, or retry the long method. The untrained control group, who did not see problems 2 to 6, tend to find the simple

routes straight away, which shows the difficulty was created by the training sequence rather than by the problems themselves: it is negative transfer from earlier success.

Question 10

Problem 5 has a shorter route that participants almost never find, while problem 7 has a much shorter one they also miss. What does the difference between those two cases suggest about what has to be true of an alternative before a set will be broken?

In problem 5 the shorter route (A plus 2C) saves only one pour and appears while the routine is still building, so it is not conspicuous enough to interrupt a method that is working. In problem 7 the alternative saves two pours, yet is still missed because the long method also works. Neither inefficiency nor the availability of a simpler route is enough to break a set while the established method still succeeds. What seems to be required is that the established method actually fails, as in problem 9, or that the alternative is made salient. A set is broken by failure, not by the mere existence of something better.

Question 11

Set out the three steps of analogical transfer. Given that a third solved the radiation problem after reading the fortress story but three-quarters solved it when told the story was relevant, what kind of failure is the main obstacle?

Analogical transfer proceeds by noticing that a source problem is relevant, mapping its parts onto the target (fortress to tumour, army to ray, mined roads to healthy tissue), and applying the mapping to generate a parallel solution (several weak beams converging on the tumour). About a tenth solved the radiation problem cold, a third after reading the fortress story, and three-quarters when told the story was relevant. Once told, most people could map and apply the analogy, so the main obstacle is noticing: a retrieval failure, not a failure to understand the story or the analogy.

Question 12

How does comparing two source problems improve transfer, and what is a problem schema? State the analogical paradox and give two explanations for it.

Comparing two analogous source stories forces out their shared deep structure and strips away the surface details on which they differ, roughly doubling transfer (Gick & Holyoak; Gentner and colleagues' analogical encoding). What is extracted is a problem schema, an abstract representation of the structure common to a class of problems, which can then be recognised in a new problem. The analogical paradox (Dunbar, 2001) is that analogical transfer is fragile in the laboratory, where people cling to surface similarity, yet scientists, politicians and engineers use analogies fluently in the real world. Explanations: expertise means the deep structure of problems in one's own domain is what is noticed first; and laboratory problems are stripped of the rich context that, in real settings, cues retrieval of the right source. (Dunbar also found that working scientists mostly draw analogies from closely related systems.)

Question 13

How do experts differ from novices in the chess-reconstruction studies, and what exactly does the random-board control rule out?

When shown a board for a few seconds, experts reconstructed real game positions far better than novices, but on randomly arranged pieces they were no better, or only slightly better in larger samples. Experts store tens of thousands of chunks, recurring meaningful configurations of pieces, so a real position can be encoded in a few chunks. The random-board control largely rules out superior general visual memory as the explanation, because better memory would help on any board. What remains is the recognition of meaningful patterns, which explains why expertise is context-bound.

Question 14

What is the difference between sorting problems by surface and by deep structure, and how does it connect expertise to analogical noticing?

Surface structure refers to the objects and features a problem mentions, such as inclined planes or springs; deep structure refers to the underlying principle it turns on, such as conservation of energy. Chi and colleagues found novices sorted physics problems by surface features and experts by deep principles. Because experts represent new problems by their deep structure, they connect them to relevant knowledge and to analogous problems already solved, which is precisely the noticing step that novices find hard in analogical transfer. Expert problem representation and expert analogical noticing are the same ability, which also helps explain the analogical paradox.

Question 15

Where would you expect an expert's mental set to be most costly? Describe a situation in which a novice might outperform an expert.

An expert's mental set is most costly when a problem presents familiar surface features but requires an unfamiliar principle or approach: the expert's rich, automatic domain knowledge immediately suggests the standard approach, which locks them into a familiar but wrong method, as in Luchins' studies. For example, a clinician might quickly diagnose a common condition from a typical-looking presentation when the case is actually something rarer that requires a different line of investigation, or an experienced engineer might apply a standard design to a problem whose constraints have changed. A novice, lacking the entrenched routine, may approach the problem fresh and notice the simpler or correct solution, much as the untrained control participants found the two-pour routes.

Question 16

Define divergent and convergent thinking, and name the four dimensions on which Unusual Uses responses are scored.

Divergent thinking is open-ended thinking that generates many possible answers, none simply right or wrong; convergent thinking narrows towards a single correct answer. Unusual (Alternative) Uses responses are scored on fluency (how many responses), originality (how rare they are), flexibility (how many distinct categories they span) and elaboration (how much detail they contain).

Question 17

Why should you be cautious about treating divergent-thinking scores as a measure of creativity? What is the gap between the task and the construct?

Divergent-thinking tasks measure the ability to produce many varied responses to a short prompt under time pressure in a laboratory. Creativity, defined as producing ideas that are original, useful and surprising, typically involves sustained, knowledge-heavy work in a real domain, evaluated by others, and requires evaluation and refinement (convergent processes) as well as generation. The correlations between divergent-thinking scores and real-world creative achievement are modest. So a divergent-thinking score is a claim about a laboratory task, not about creative work, and claims that a person or group is "more creative" should be checked against what was actually measured.

Question 18

Summarise the evidence that the default mode and executive control networks both contribute to creative cognition, and explain why their cooperation is surprising.

Ellamil and colleagues had art students alternate between generating and evaluating ideas for book covers in the scanner: generation recruited medial temporal memory regions, while evaluation recruited executive and default-mode regions together, and the two networks were positively coupled throughout. Mayseless and colleagues found that more original Alternative Uses responses were associated with greater activity in default-mode regions such as medial prefrontal and posterior cingulate cortex. Beaty and colleagues found the default

mode network coupling with salience regions and later with executive regions during divergent thinking. Their cooperation is surprising because the default mode network (active during rest and self-generated thought) and the executive control network (active during focused goal-directed control) are normally anti-correlated: states that engage one usually suppress the other.

Question 19

Why is the pre-stimulus finding in the prepared-mind studies more surprising than the gamma burst at the moment of insight?

The gamma burst over right anterior temporal cortex about a third of a second before an insight solution is reported is a correlate of the solution itself, so it is not surprising that solving by insight has a neural signature at that moment. The pre-stimulus finding is that brain activity in the seconds before the problem even appears predicts how it will be solved: more frontal and midline activity precedes insight solutions, more occipital activity precedes analytic ones. The state of the brain going into a problem biases the route to its solution before there is any problem to solve, which echoes the anticipatory preparation of Chapter 2.

Question 20

The default mode and executive control networks are normally anti-correlated, yet both contribute to creative cognition. Explain why that raises a question about mechanism, and what role the salience network is proposed to play. What direct evidence supports the proposal, and why is it still not settled?

If the default mode and executive control networks normally suppress each other, there must be some mechanism that lets them work together during creative cognition rather than competing. The triple-network model proposes that a third network, the salience network and particularly the anterior insula, acts as a switch, engaging the executive network and standing down the default mode network when something salient arrives, so flexible creative thought might depend on how well switching between them is controlled. Direct support: during divergent thinking the default mode network couples first with salience regions, including the insula, and later with executive regions (Beaty et al., 2015). It is not settled because the triple-network model was developed to explain clinical conditions rather than creativity, and the evidence that switching causes creative cognition is limited.

Question 21

Give the four standard explanations for why group brainstorming underperforms individuals working alone, and say which remedy addresses which mechanism.

Production blocking: only one person can speak at a time, so others must hold ideas in limited working memory while waiting and often lose them; generating alone first and pooling afterwards removes it. Evaluation apprehension: people withhold ideas that might look foolish, which selectively removes the most original contributions; collecting ideas anonymously or in writing reduces it. Social loafing: contributions are not individually identifiable, so effort falls; making each person's contributions identifiable to the facilitator, even if anonymous to the group, reduces it. Conformity and anchoring: early ideas set a direction that later ideas cluster around, convergence arriving too early; writing ideas independently before sharing them removes anchoring on whoever speaks first. More generally, separate the generation phase from the evaluation phase.

Question 22

Barack and colleagues (2024) found that higher attention-deficit traits predicted more exploration and higher reward rates in a foraging task. Explain precisely why this does not establish that the traits were evolutionarily selected for, and what evidence would be needed.

The finding is a correlation between traits, measured by screening rather than diagnosis, and performance in a modern computerised task. It shows there are environments in which these traits pay off. Selection for a trait is a claim about ancestral environments and differential reproductive success over many generations, which a modern laboratory task cannot reveal, and ancestral selection pressures leave no direct record. "This trait helps in this task" and "this trait was selected for" are different claims. Supporting the evolutionary claim would

require evidence about the genetic basis of the traits, their fitness consequences in ancestral-like environments, and signatures of selection on the relevant genes, which is extremely hard to obtain.

Question 23

A claim that a group is “more creative” rests on divergent-thinking scores. Set out two distinct reasons for caution, one about the sample and one about the measure.

The sample: many studies use subclinical groups, people with elevated traits rather than a clinical diagnosis, or self-selected samples, so results may not generalise to the group the claim refers to, and the advantage may be specific to particular domains or motivational conditions. For ADHD, Hoogman and colleagues found divergent thinking somewhat elevated in subclinical groups, convergent thinking not enhanced. The measure: divergent-thinking scores measure producing many varied responses to a short prompt under time pressure, which predicts real-world creative achievement only modestly. So a higher score supports a claim about a laboratory task, not about creative work in a real domain.

Chapter 11: Judgement, Decisions and Reasoning

Question 1

Distinguish inductive from deductive reasoning, and name the three factors that determine the strength of an inductive argument.

Deductive reasoning starts from general premises and derives a conclusion that is guaranteed to be true if the premises are true. Inductive reasoning starts from specific observations and generalises to a conclusion that is probable but never guaranteed: a single black swan overturns "all swans are white". The defining difference is certainty rather than direction. The strength of an inductive argument depends on the representativeness of the observations (are they typical of the whole, or a skewed sample?), the number of observations, and the quality of the evidence (precise, well controlled and from good sources rather than casual impressions).

Question 2

Explain the difference between validity and truth, using the "all flowers are animals" syllogism. Why is belief bias evidence that we judge form by content?

Validity concerns form: a syllogism is valid if the conclusion follows from the premises, so that if the premises were true the conclusion would have to be true. Truth concerns content: whether the premises actually hold. "All flowers are animals; all animals can jump; therefore all flowers can jump" has false premises and an absurd conclusion, yet is valid, because the conclusion follows from the premises. Belief bias is the tendency to judge a syllogism valid when its conclusion is believable and invalid when it is not, regardless of the logic: many people reject the valid "some bananas are pink" conclusion because it is unbelievable. Since validity depends only on form, judging it by believability shows people are evaluating content when they are supposed to evaluate form.

Question 3

Using Figure 11.1, explain why "all mammals are animals; all dogs are animals; therefore all dogs are mammals" is invalid even though its conclusion is true. What is the general method for showing a syllogism invalid?

The premises place both dogs and mammals inside the category of animals but say nothing about how dogs and mammals relate to each other. As the Euler circles on the right of Figure 11.1 show, the premises are equally consistent with an arrangement in which the dogs circle sits entirely outside the mammals circle, both inside animals. In that arrangement the conclusion fails, so it does not follow from the premises, even though in reality dogs are mammals and the conclusion is true. The general method is to find one arrangement consistent with all the premises in which the conclusion is false; you do not need to show the conclusion is actually false.

Question 4

For conditional syllogisms, which two forms are valid and which two are not? Give the invalid form that reasons "the street is wet, therefore it is raining", and name it.

Valid: modus ponens (if p then q; p; therefore q) and modus tollens (if p then q; not q; therefore not p). Invalid: affirming the consequent (if p then q; q; therefore p) and denying the antecedent (if p then q; not p; therefore not q). "If it rains the street gets wet; the street is wet; therefore it is raining" is affirming the consequent, invalid because the street could be wet for other reasons.

Question 5

Why do people fail the abstract Wason selection task but succeed with the drinking-age version? Give both explanations, and say what the contrast has in common with the analogical paradox of §10.7.

In the abstract task (if vowel then even number, with cards E, K, 4 and 7), the correct answer is E and 7, since a vowel behind 7 would break the rule, the modus tollens move. Most people choose E and 4, and only about 4

per cent get it right. With the drinking-age rule (if drinking beer, over the legal age), about three-quarters correctly turn the beer card and the under-age card, although the logic is identical. One explanation is the permission schema: a lifetime of practice with rules of the form "to do X you must satisfy Y" guides the search. The other is an evolved cheater-detection ability tuned to spotting people who take a benefit without paying the cost. Like the analogical paradox, the contrast shows that abstract laboratory versions are hard while meaningful real-world versions of the same structure are easy: performance depends on the fit between the problem's presentation and the machinery brought to it.

Question 6

Distinguish the availability and representativeness heuristics, and give an example of base-rate neglect.

The availability heuristic judges frequency or likelihood by how easily examples come to mind; the representativeness heuristic judges whether something belongs to a category by how much it resembles the category stereotype. Availability misleads when ease of recall reflects vividness or media coverage rather than frequency, as when snakebite seems deadlier than choking. Base-rate neglect: told that Steve is shy, tidy and detail-oriented, people judge him more likely to be a librarian than a farmer because he fits the librarian stereotype, ignoring the fact that farmers vastly outnumber librarians, so a random quiet man is still probably a farmer.

Question 7

Why can the conjunction fallacy never be a correct probability judgement? Explain in terms of what adding detail does to resemblance and to probability.

The conjunction rule of probability states that the probability of two things being true together can never exceed the probability of either alone, because the conjunction is a subset of each component (feminist bank tellers are a subset of bank tellers). Adding detail, such as "and a feminist", can make a description resemble the person more, raising its representativeness, but every added condition can only keep the probability the same or reduce it. Judging the conjunction more probable follows resemblance instead of probability, and because of the subset relation it can never be correct.

Question 8

What does expected utility theory predict, and how do the Denes-Raj and Epstein jelly-bean results depart from it? Why does the fact that participants understood the probabilities make the result stronger evidence rather than weaker?

Expected utility theory predicts that a rational chooser selects the option with the highest expected utility, here the small bowl, which gives a 10 per cent chance of winning against 7 per cent (or 9 per cent) in the large bowl. Many participants chose the large bowl with 7 red beans in 100, and a majority did when it held 9 in 100: the ratio bias, or denominator neglect. That they knew the probabilities were against them, and still reported feeling they had a better chance with more winning beans in view, makes the result stronger evidence. If they had misunderstood the arithmetic, the result would show only a calculation error. Because the arithmetic was understood and overridden, two processes must be competing for the decision, an intuitive one responding to the number of winners and a deliberate one that knows the odds, which supports the dual-system account.

Question 9

Explain the framing effect in terms of risk attitudes for gains versus losses, and give a real-world example of choice architecture exploiting status-quo bias.

The same choice framed in terms of gains ("saves 200 people") makes people risk-averse, so they choose the sure option; framed in terms of losses ("400 people will die"), it makes them risk-seeking, so they choose the gamble to avoid a certain loss. This reversal reflects a value function that is concave for gains and convex for losses. Choice architecture exploiting status-quo bias: organ-donation rates are far higher in countries with an opt-out system, where you are a donor unless you decline, than in opt-in systems, because people tend to stick with the default; the same applies to automatic pension enrolment.

Question 10

Set out the homeostatic account of the difference between emotions and feelings, and explain how feelings could improve rather than degrade a decision.

Every organism must keep its internal state within the narrow range that sustains life (homeostasis). On this account, emotions are the bodily responses that serve that regulation, such as changes in heart rate, gut and posture, while feelings are the conscious perception of how that regulation is going, a report on the state of the organism (Damasio, 2018). Because feelings track whether things are going well or badly for the body, they attach a value to experiences and anticipated outcomes, a rough tag of good-for-me or bad-for-me. That tag lets options be evaluated quickly without computing each one through, and lets the future consequences of actions weigh on present choices.

Question 11

Bechara and colleagues (1994) found their patients were guided by immediate prospects only. Explain why that impairs decision-making rather than making them more rational, and what it suggests about the relationship between emotion and reason.

In the Iowa Gambling Task two decks give large immediate rewards but larger occasional penalties and lose money over time, while two give smaller rewards and come out ahead. Controls learned to favour the profitable decks; patients with ventromedial prefrontal damage did not, despite preserved intellect, and in their lives they repeatedly decided against their own interests. Being guided by immediate prospects only means ignoring future consequences, which is not rational: a rational chooser should weigh long-run outcomes. What the damage removed was the feeling-based value signal that makes the future weigh on the present. Emotion is therefore not the opposite of reason but part of what makes good decisions possible.

Question 12

Kermer and colleagues found the loss-gain asymmetry in forecasts but not in experience. Explain what that implies about loss aversion, and what mechanism the authors identified.

Forecasters predicted that losing a gamble would hurt more than an equivalent win would please, even for a favourable gamble (50 per cent chance of winning \$5 against losing \$3), but those who actually lost were no less happy than those who won, and loss forecasters overestimated their unhappiness. Loss aversion is therefore at least partly an affective forecasting error: it operates on anticipated losses rather than on how losses are actually felt. The mechanism was that people failed to anticipate how readily they would rationalise a loss, for example reframing losing \$3 of \$5 as still having made \$2; they overestimate how much they will dwell on a bad outcome and underestimate how quickly they will make peace with it.

Question 13

Describe Sanfey and colleagues' ultimatum-game findings. What does anterior-insula activity predict, and in what sense does rejecting an unfair offer stop looking irrational once repeated interactions are considered?

In the ultimatum game a proposer splits a sum and the responder accepts, so both are paid, or rejects, so neither is paid. A purely payoff-maximising responder should accept any non-zero offer, but people routinely reject unfair offers. Sanfey and colleagues found unfair offers activated the anterior insula (associated with negative emotional states such as disgust), the dorsolateral prefrontal cortex (cognitive control) and the anterior cingulate (conflict detection). The stronger the anterior-insula response to an unfair offer, the more likely the person was to reject it, while prefrontal activity favoured accepting the money. Rejection is irrational only if the immediate payoff is the whole story. In repeated interactions, a reputation for refusing to be exploited has value, because it deters future unfair offers, so the behaviour can be well adapted outside the one-shot laboratory game.

Question 14

State Gigerenzer's ecological-rationality argument. Why does the natural-frequencies result relocate part of the fault away from the reasoner?

Gigerenzer argues that heuristics are not bugs but fast and frugal tools tuned to the environments in which they evolved and are used. Judged against abstract formal standards they look like failures, but judged by whether they produce good decisions quickly with limited information and time they often do very well, and simple rules can match or beat complex models on real problems. Many biases shrink or disappear when problems are posed in natural frequencies ("10 out of every 1,000 women") rather than probabilities ("0.01"), including base-rate neglect in medical diagnosis, even among practising doctors. If the error depends on the format of the question, part of the fault lies in the format rather than in the mind, which may be well suited to frequencies of observed events, the form in which information was naturally encountered.

Question 15

Using Table 11.1, name two effects from this chapter that are well replicated and two claims from the wider literature that are not. What distinguishes the two groups?

Robust and well replicated: framing effects; base-rate neglect and the conjunction fallacy; belief bias in syllogistic reasoning; confirmation and myside bias; classic anchoring (any two). Not holding up or smaller than first claimed: ego depletion, which failed to replicate in a multi-laboratory pre-registered test ($d = 0.04$, 95% CI from -0.07 to 0.15); many social-priming effects, such as cleanliness cues altering moral judgement; and loss aversion, which is contested (any two of these). The robust effects tend to have a clear mechanism and a large, replicable behavioural signature. Those that shrank tend to be ones inferred from a small difference on a single measure.

Question 16

"People are fundamentally irrational." Evaluate this claim using the material in §11.8, and say what a more defensible summary would be.

The claim is too strong. The biases are real, and their direction is predictable, which is what makes them correctable, but many are smaller and more context-dependent than popular accounts suggest, and some (ego depletion, several social-priming effects) have not held up. Many errors shrink when information is presented in natural frequencies, so they partly reflect the format of the problem, and abstract tasks such as the Wason task are failed while meaningful versions with the same logic are solved. Heuristics produce good decisions quickly with limited information in real environments, and behaviour that looks irrational in a one-shot laboratory task, such as rejecting unfair offers, may be adaptive in the world. A more defensible summary is that human judgement is a system built to decide quickly and adequately with incomplete information, which mostly succeeds, and that rationality is a relation between a mind and its environment: good reasoning means recognising the specific situations where intuition reliably misleads and deliberately engaging slow thinking there.

Chapter 12: Drugs, Disorders and the Cognitive Brain

Question 1

A study finds that habitual coffee drinkers perform better after caffeine than after placebo. Why is that not yet evidence that caffeine enhances cognition, and what design would make it so?

Habitual coffee drinkers tested after a night without caffeine are not at a neutral baseline; they are in mild withdrawal (tiredness, poorer concentration, headache). A dose of caffeine that removes the withdrawal will improve their performance relative to placebo even if it never lifts them above where they would be without the habit. This is withdrawal reversal. To show genuine enhancement you need a comparison with non-users or low users, or with habitual users who have not been deprived, tested on caffeine and placebo under double-blind conditions. Rogers and colleagues (2013) did this and found caffeine restored withdrawn consumers to normal, but did little for the alertness or performance of non-consumers beyond speeding motor responses: faster, but not smarter.

Question 2

Explain the difference between an agonist, an antagonist and a positive allosteric modulator, giving one drug example of each from this chapter.

An agonist binds to a receptor and activates it, as nicotine does at nicotinic acetylcholine receptors (opioids at opioid receptors, and psilocybin and LSD at 5-HT_{2A} receptors, are further examples). An antagonist binds without activating, blocking the natural transmitter, as caffeine does at adenosine receptors (naloxone at opioid receptors is another). A positive allosteric modulator binds to a different site and amplifies the response to transmitter already present, as benzodiazepines do for GABA at GABA-A receptors.

Question 3

Why does regular caffeine use lead to withdrawal symptoms? Refer to what happens to adenosine receptors.

Caffeine is an antagonist at adenosine receptors, blocking the sleepiness-promoting signal of adenosine. With regular exposure, cells compensate by expressing more adenosine receptors. When caffeine is withdrawn, as after a night without it, this enlarged receptor population is fully exposed to normal levels of adenosine, so the adenosine signal is stronger than usual, producing tiredness, poorer concentration and headache. The withdrawal symptoms are roughly the mirror image of the drug's effects.

Question 4

Using the inverted U, explain why a stimulant might help one student and impair another.

Dopamine's effects on working memory and cognitive control follow an inverted U: performance is best at an intermediate level of dopamine signalling, and both too little and too much impair it. People differ in where they start. A student who starts below the optimum, for example one with low working-memory capacity, may be moved by a stimulant towards the peak and improve. A student who starts near the optimum may be pushed past it by the same dose, so their performance falls. Averaging across such people can make a drug look as if it does nothing.

Question 5

Bowman and colleagues (2023) found that “smart drugs” increased effort but reduced its quality. Explain how both can be true at once, and why the result matters for students.

Effort and the quality of effort are different things. Bowman and colleagues found methylphenidate, dexamfetamine and modafinil all made participants spend longer and make more moves on knapsack problems, but the value of their solutions fell, apparently because their search became more random and less systematic.

Stimulants change motivation (tasks feel more interesting and effort cheaper) more than capacity. Differences between participants also shrank and reversed, so people who did well on placebo tended to do worse on the drugs, consistent with the inverted U. For students this means the feeling of working harder on a non-prescribed stimulant is not evidence of working better, and those most tempted to take such drugs to do even better may have the most to lose.

Question 6

According to the alcohol myopia account, why might a drink make one person calmer and another more anxious?

Alcohol myopia (Steele & Josephs, 1990) holds that alcohol narrows attention to the most salient cues in the immediate situation, while quieter, more abstract cues that take capacity to keep in mind drop out. Alcohol therefore has no fixed emotional effect: its effect depends on where attention is directed. A drinker left alone with a worry has nothing to distract them, so the worry dominates the narrowed window and they become more anxious. A drinker absorbed in a game or conversation has the worry pushed out of the window, so they feel calmer.

Question 7

Why can a person in an alcohol-related blackout hold a conversation they will never remember? Which stage of memory is affected?

Alcohol primarily interferes with forming new long-term memories while leaving previously established memories and short-term maintenance intact (White, 2003). In a blackout the person can keep information active long enough to hold a conversation, find their way home or drive (which they should never do), but the information is never transferred into long-term storage, so it cannot later be retrieved. The stage affected is encoding (consolidation into long-term memory), not retrieval, linked to disrupted hippocampal activity, a pharmacological echo of anterograde amnesia.

Question 8

What would a confound have to look like to produce the Dunedin cannabis result?

The confound would need to be present before cannabis use began, or at least not be a consequence of it. It would have to predict a decline in IQ relative to each person's own childhood score, not just a low IQ. It would have to grow stronger with more persistent use, to reproduce the dose-response relation. And it would have to survive adjustment for childhood socioeconomic status, low childhood self-control and family history of substance dependence. Candidates include early-emerging mental health problems, disrupted education or chronic poor sleep, if they both lead to persistent use and cause cognitive decline.

Question 9

Jackson and colleagues (2016) found no greater IQ decline in cannabis-using twins than in their non-using co-twins. What does this design control for that a birth-cohort study does not, and what can it not control for?

Comparing a twin who used cannabis with a co-twin who did not controls for everything the twins share: their family environment (upbringing, socioeconomic status, schooling) and all or much of their genes. A birth-cohort study can adjust only for the factors it measured. The co-twin design cannot control for factors that differ between twins, such as different peer groups, mental health problems or experiences affecting only one twin, and its estimates become uncertain when few pairs differ sharply in use. In Jackson and colleagues' adolescent samples, use was mostly light and relatively short, so the design cannot rule out effects of the very heavy, prolonged use seen in the Dunedin long-term users.

Question 10

What does the REBUS model propose that psychedelics do, and how does it connect to the top-down and bottom-up processing discussed in Chapter 8?

REBUS (relaxed beliefs under psychedelics; Carhart-Harris & Friston, 2019) proposes that the brain constantly predicts its input, each prediction carrying a precision that determines how strongly it resists contrary evidence, and that psychedelics relax the precision of high-level prior beliefs. Bottom-up information then has more influence than usual. Perception becomes less constrained by expectation, which could explain vivid distortions, and deeply held beliefs about self and world become temporarily open to revision, the basis of the claimed therapeutic effect. This connects to the view in Chapter 8 that perception is a negotiation between top-down expectations and bottom-up sensory evidence: psychedelics shift the balance towards bottom-up evidence.

Question 11

Why is it so hard to run a properly blinded trial of a psychedelic drug, and why does it matter?

Within an hour of taking psilocybin nearly everyone knows whether they have had it, and so usually do the therapists in the room, so the double blind breaks down. Once participants know their group, expectations can do much of the work: those who hoped for a breakthrough and received the drug may report improvement because they expect it, while those who realise they had placebo may be disappointed and report feeling worse. The comparison group is then not what it appears to be, so a difference between groups may reflect expectancy rather than the drug. This was among the concerns that led the US regulator to decline MDMA-assisted therapy for PTSD.

Question 12

Lim and Dinges (2010) found that simple attention suffers more from sleep deprivation than reasoning does. Give one explanation of this counterintuitive result.

Monotonous, simple tasks provide nothing to counteract mounting sleep pressure, so attention drifts and brief lapses, sometimes actual microsleeps, intrude. A complex, engaging reasoning task recruits effort and interest that can temporarily hold those lapses off. Simple sustained attention therefore suffers most, which matters because everyday tasks that depend on it, such as night driving on an empty road or monitoring a screen, are those where a single lapse can be catastrophic.

Question 13

What does general anaesthesia suggest about the neural basis of consciousness? Link your answer to §2.8.

Anaesthetics act on varied molecular targets (propofol enhances GABA inhibition, ketamine blocks glutamate receptors) yet converge on unconsciousness, and unconsciousness does not require the cortex to fall silent: early sensory areas can still respond to stimulation. What appears to fail is integration, the communication between cortical areas and between cortex and thalamus that binds local processing into a unified experience. Ketamine also shows that responsiveness and experience can come apart. This fits §2.8, where awareness was said to depend on reciprocal, synchronised exchange across brain regions rather than on activity in any one area, as in recurrent-processing and global-workspace accounts.

Question 14

Compare the memory impairment typical of Alzheimer's disease with the cognitive impairment typical of Parkinson's disease.

In Alzheimer's disease the earliest and most prominent deficit is in episodic memory, with rapid forgetting: learned information is lost abnormally quickly over a delay, and cues and recognition help little. This points to a failure of consolidation and storage, consistent with early hippocampal and medial temporal involvement; semantic, executive and visuospatial deficits follow as the disease spreads. In Parkinson's disease the core cognitive deficits are fronto-executive: cognitive flexibility, planning, working memory and learning from

reinforcement, driven by disrupted dopamine signalling in striatal-prefrontal circuits. Memory problems there affect retrieval and strategic aspects more than storage, and a subgroup develops dementia.

Question 15

L-dopa improved task switching but impaired reversal learning in the same patients with Parkinson's disease (Cools et al., 2001). Use the inverted U and the uneven loss of dopamine to explain why.

Dopamine loss in Parkinson's disease is uneven: it begins in the dorsal striatum and spares the ventral striatum until later. Under the inverted U, both too little and too much dopamine impair cognition. L-dopa restores dopamine in the depleted dorsal striatum, moving those circuits towards their optimum and improving task switching, which depends on them. The same dose overdoses the relatively intact ventral striatum, pushing it past its optimum and impairing probabilistic reversal learning, which depends on it. One medication therefore helps one function and harms another in the same person on the same day.

Question 16

Distinguish "wanting" from "liking". How does the incentive-sensitisation theory explain why a person might crave a drug they no longer enjoy?

Liking is the pleasurable impact of a reward; wanting is its motivational pull, the attraction of the reward and of the cues that predict it. Incentive-sensitisation theory (Robinson & Berridge, 1993) holds that dopamine supports wanting rather than liking. In some people repeated drug use makes the wanting system increasingly, perhaps permanently, hypersensitive to the drug and its associated cues, while the liking system is not sensitised. Wanting can therefore grow even as liking fades, so a person may crave a drug intensely while enjoying it less than ever, and cues such as places, people, equipment or times of day can trigger craving on their own.

Question 17

Why might a dose that a person has taken many times become dangerous in an unfamiliar setting? What does this suggest about the word "overdose"?

Part of tolerance is learned through classical conditioning: cues that reliably precede a dose, such as a particular room or ritual, come to trigger a compensatory response that opposes the drug's effect before it arrives. In an unfamiliar setting those cues are absent, so the compensatory response does not occur and the body is unprepared for a dose it would usually tolerate, which can be fatal. Rats given a large heroin dose in a new environment were less likely to survive than those dosed in the usual one, and case reports describe the same in people. "Overdose" can therefore mislead: a dangerous dose is not simply a fixed amount but depends on the person's state and on the setting in which the drug is taken.

Question 18

Szigeti and colleagues (2021) had microdosers blind themselves. What problem does this design solve, and which problem from the blinding box in §12.6 might remain?

It solves the problem of placebo control in a real-world practice that cannot easily be studied in the laboratory. Participants who were already microdosing packed their own doses and empty placebo capsules into identical opaque capsules and QR-coded envelopes, so they did not know which they were taking in a given week but the researchers could later decode it. This allowed a placebo-controlled comparison without supplying the drug, and showed that reported improvements in well-being and cognition occurred on placebo too. What might remain is the blinding problem of §12.6: if participants could detect when they had taken a real microdose from subtle effects, they could partly break the blind, and their expectations could then contribute to the reported effects.